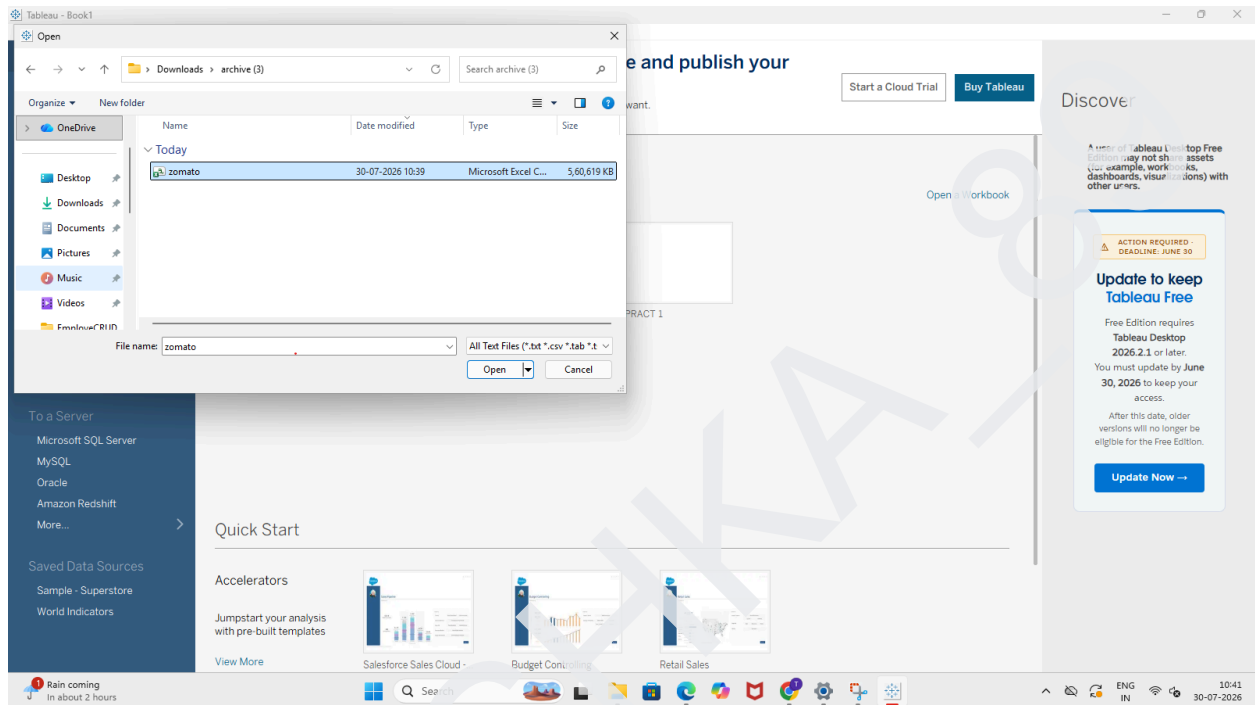


SHETH L.U.J. AND SIR M.V. COLLEGE
SUB:-Data Visualization with Power BI and Tableau
PRACTICAL NO.:-5

AIM: Analyzing Data using Table Calculations

A: Apply quick table calculations (running total, percent of total)

1. Launch Tableau. Under the Connect panel, click on Text File, select your csv, and click Open.



Create a Running Total of Votes by City

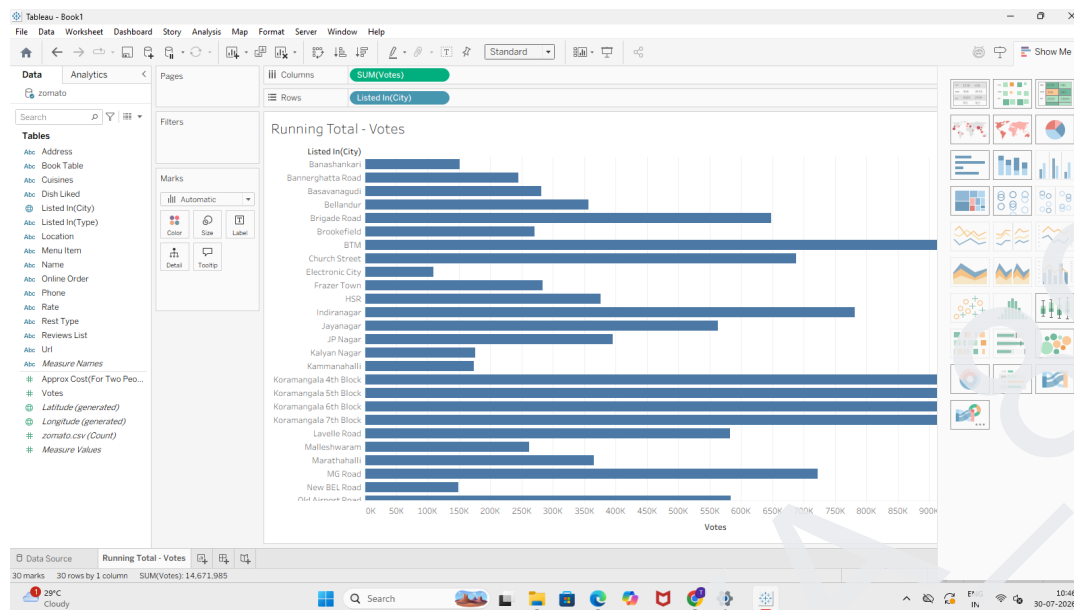
From the Data Pane, drag listed_in(city) to the Rows shelf.

Drag votes to the Columns shelf. (Tableau will automatically aggregate it as SUM(votes)).

SHETH L.U.J. AND SIR M.V. COLLEGE

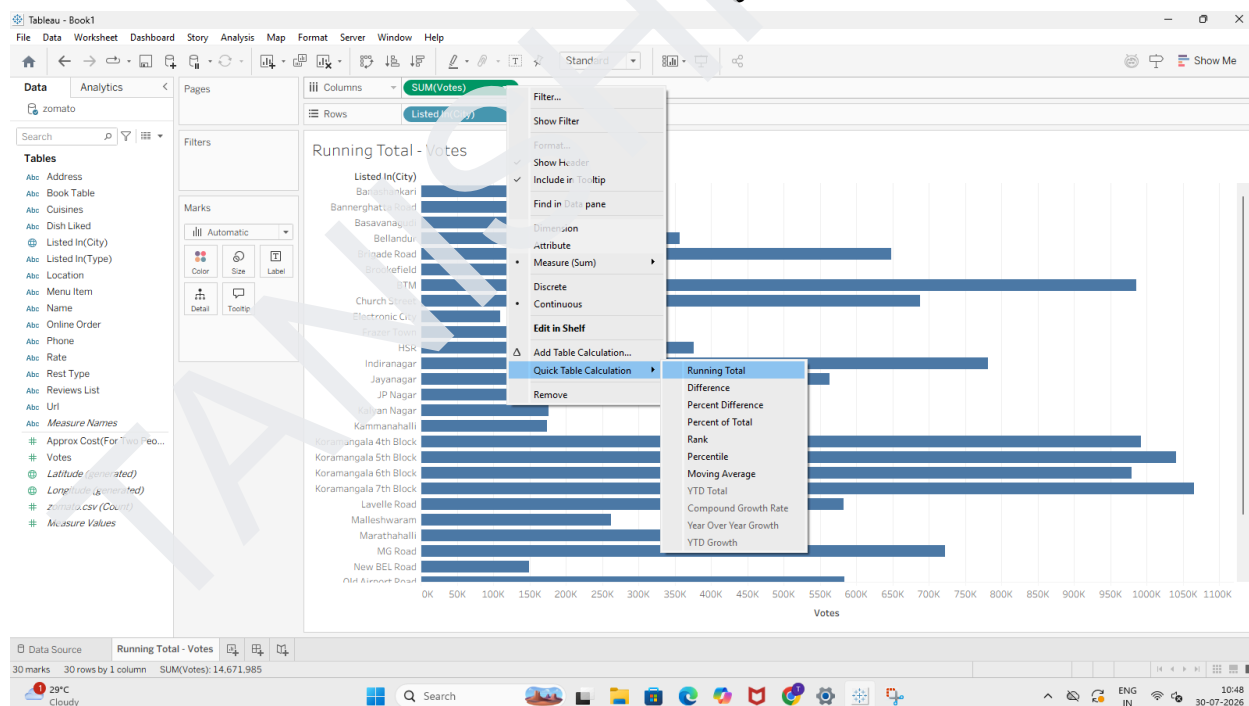
SUB:-Data Visualization with Power BI and Tableau

PRACTICAL NO.:-5



Convert to Quick Table Calculation:

- o Right-click the SUM(votes) pill on the Columns shelf.
- o Hover over Quick Table Calculation and select Running Total.



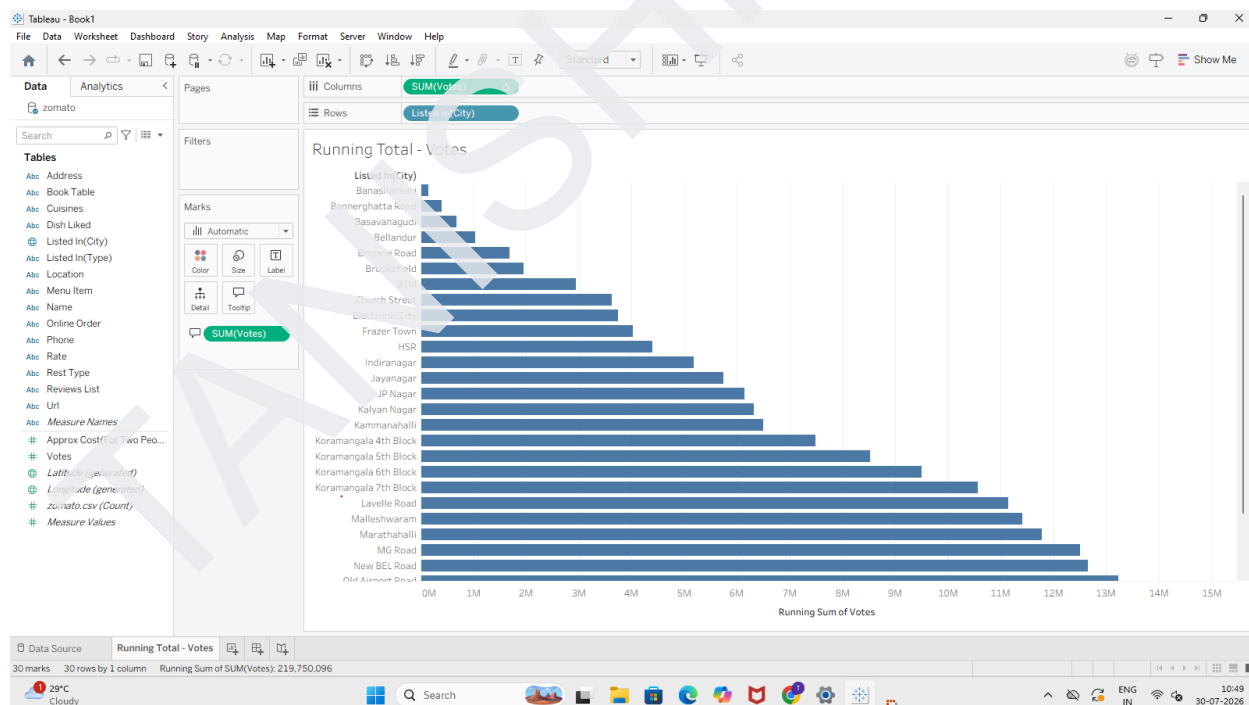
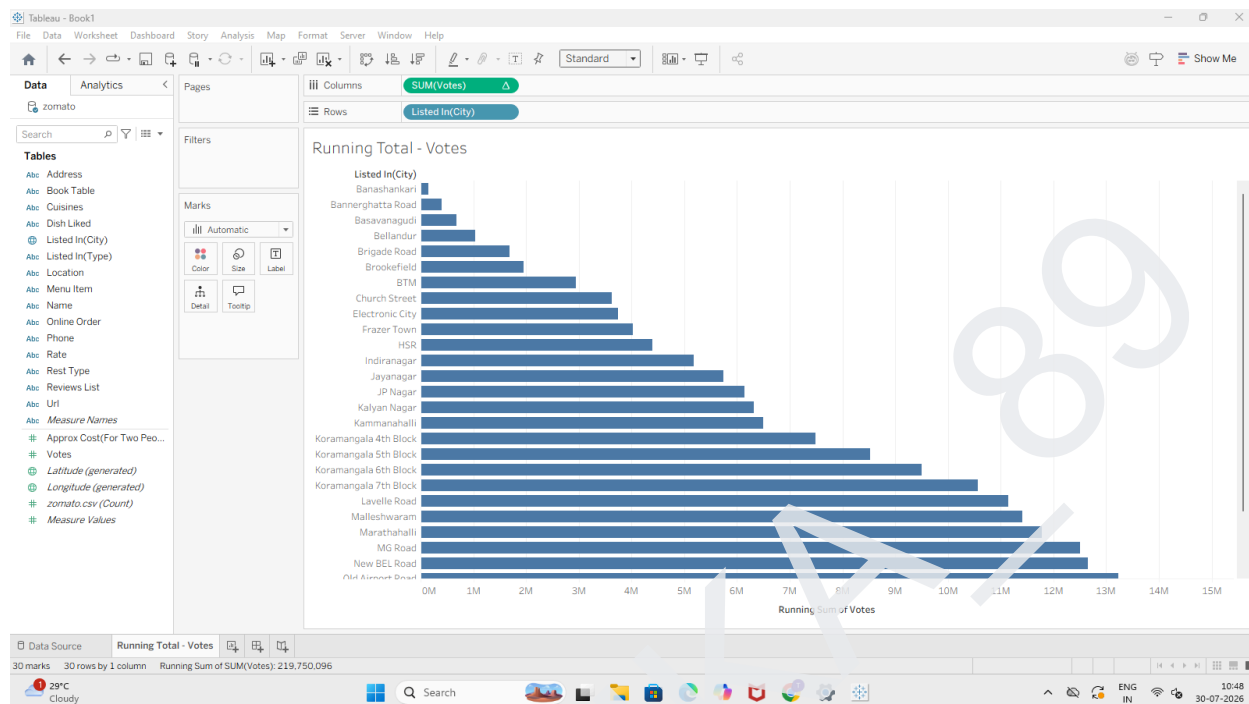
Add Visual Context (Secondary Measure)

NAME : TANISHKA KUTWAL
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SUB:-Data Visualization with Power BI and Tableau

PRACTICAL NO.:-5



Create a Percent of Total across Categories

Open a new worksheet and rename it: Percent of Total - Service Type.

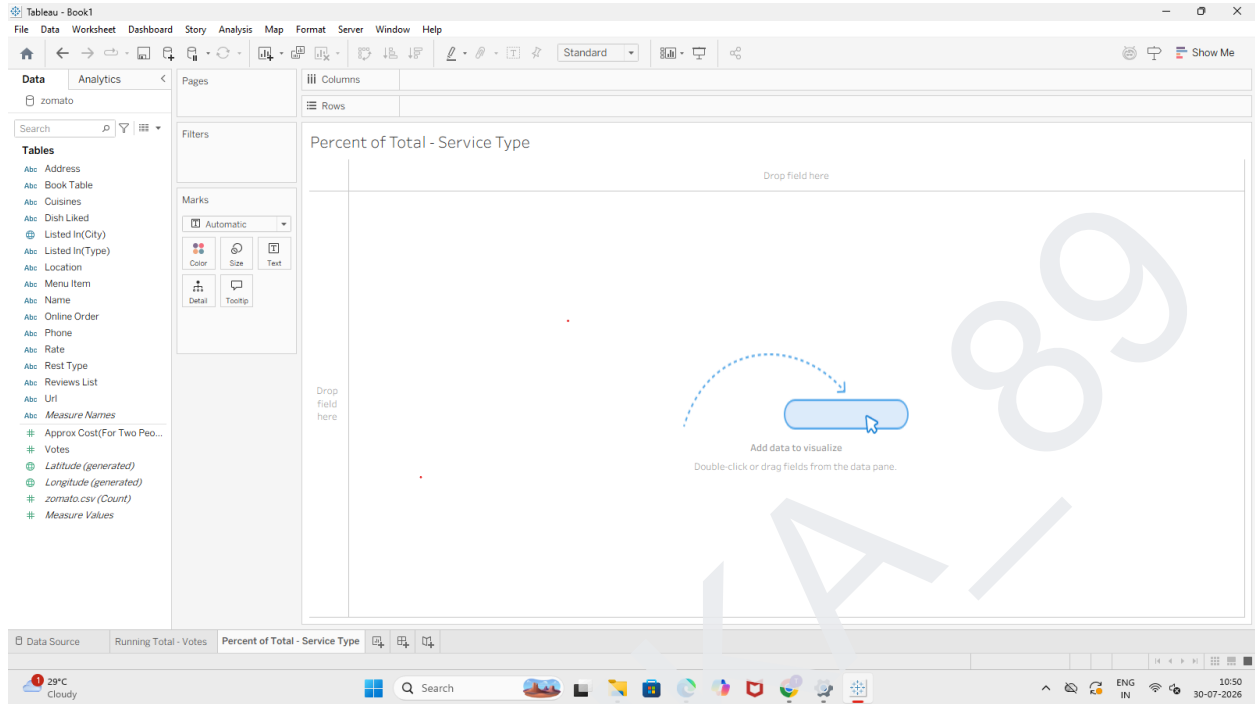
NAME : TANISHKA KUTWAL

ROLL NO.:- T089

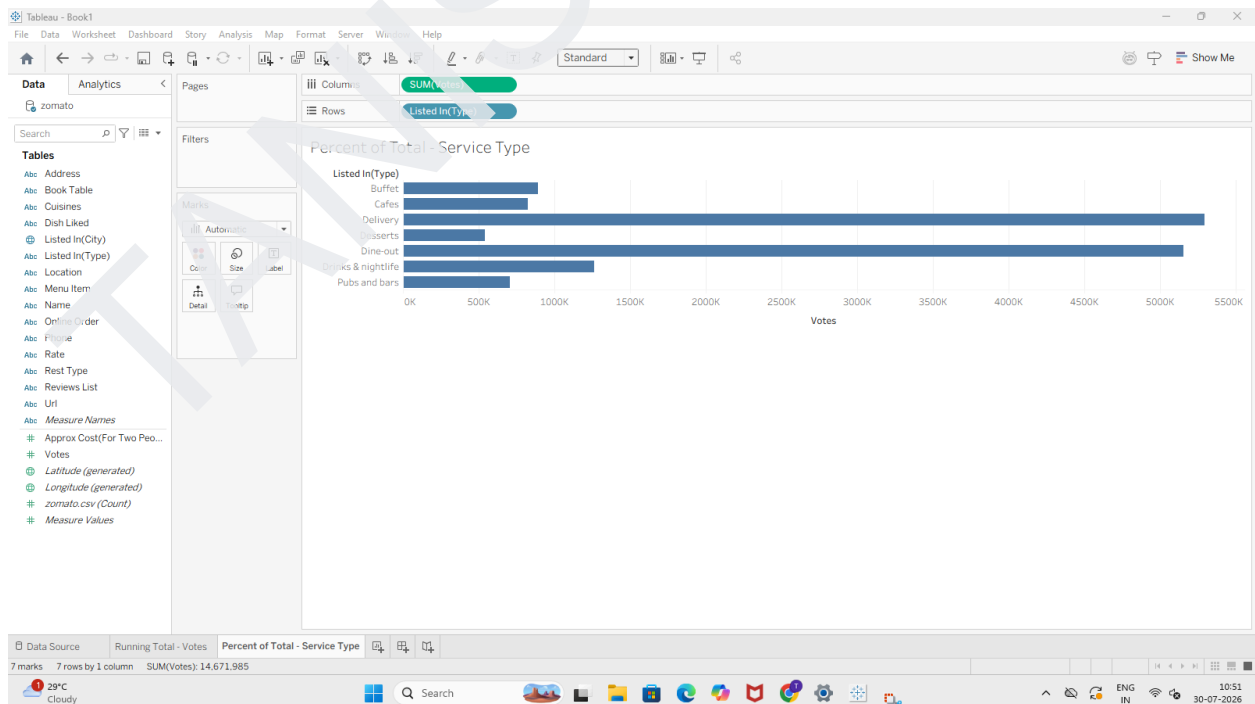
SHETH L.U.J. AND SIR M.V. COLLEGE

SUB:-Data Visualization with Power BI and Tableau

PRACTICAL NO.:-5



Drag listed_in(type) (e.g., Buffet, Cafes, Delivery, Dining) to the Rows shelf.
Drag votes to the Columns shelf.



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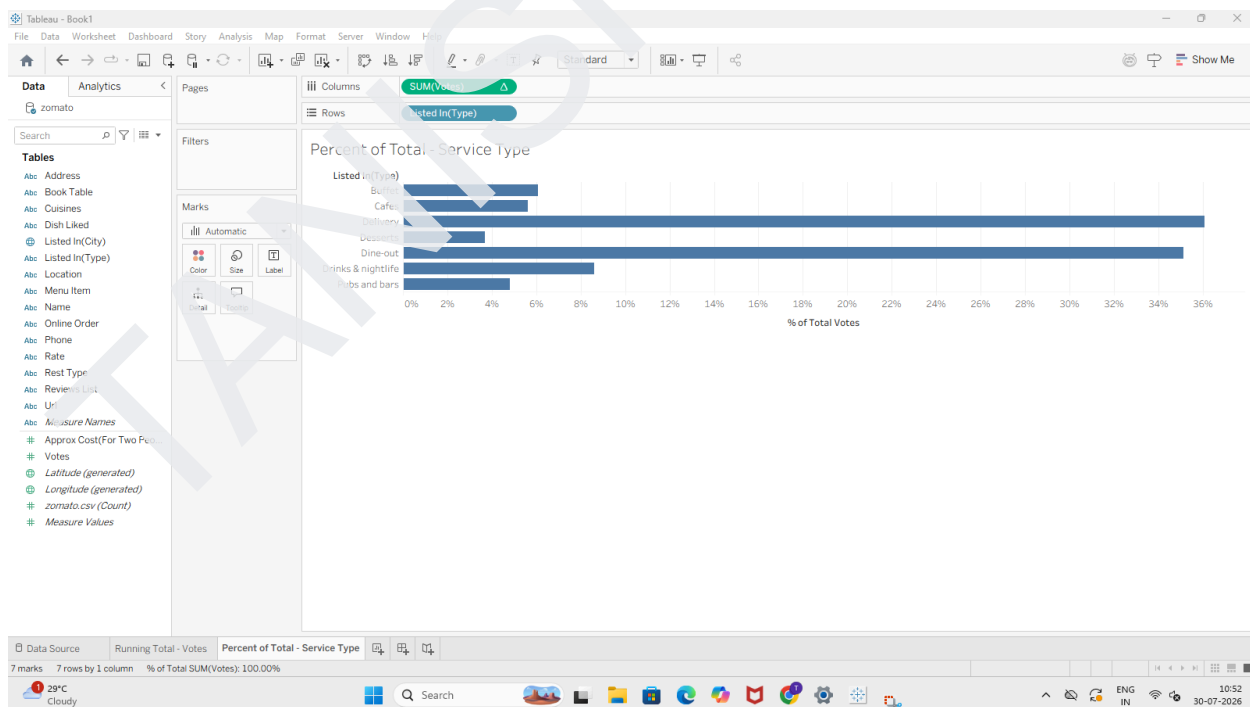
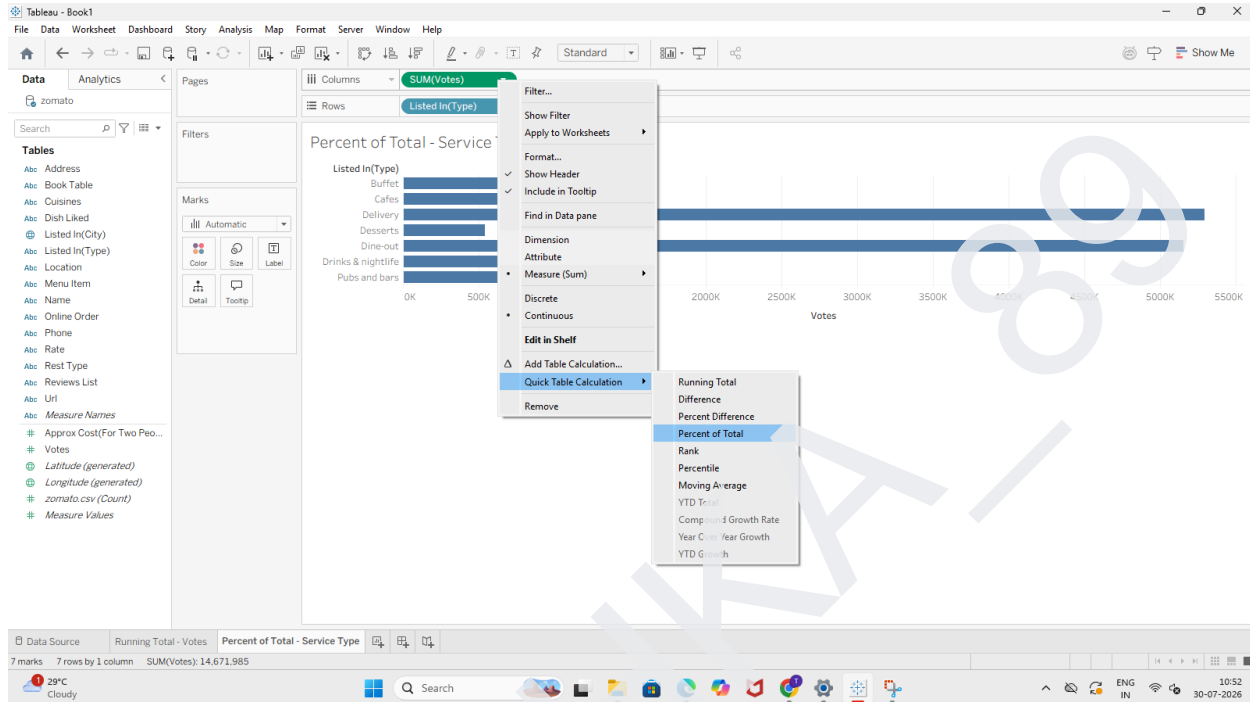
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SUB:-Data Visualization with Power BI and Tableau

PRACTICAL NO.:-5

Right-click the SUM(votes) pill on the Columns shelf.

Hover over Quick Table Calculation and select Percent of Total.



Format as Percentage:

o Right-click the SUM(votes) pill again and select Format...

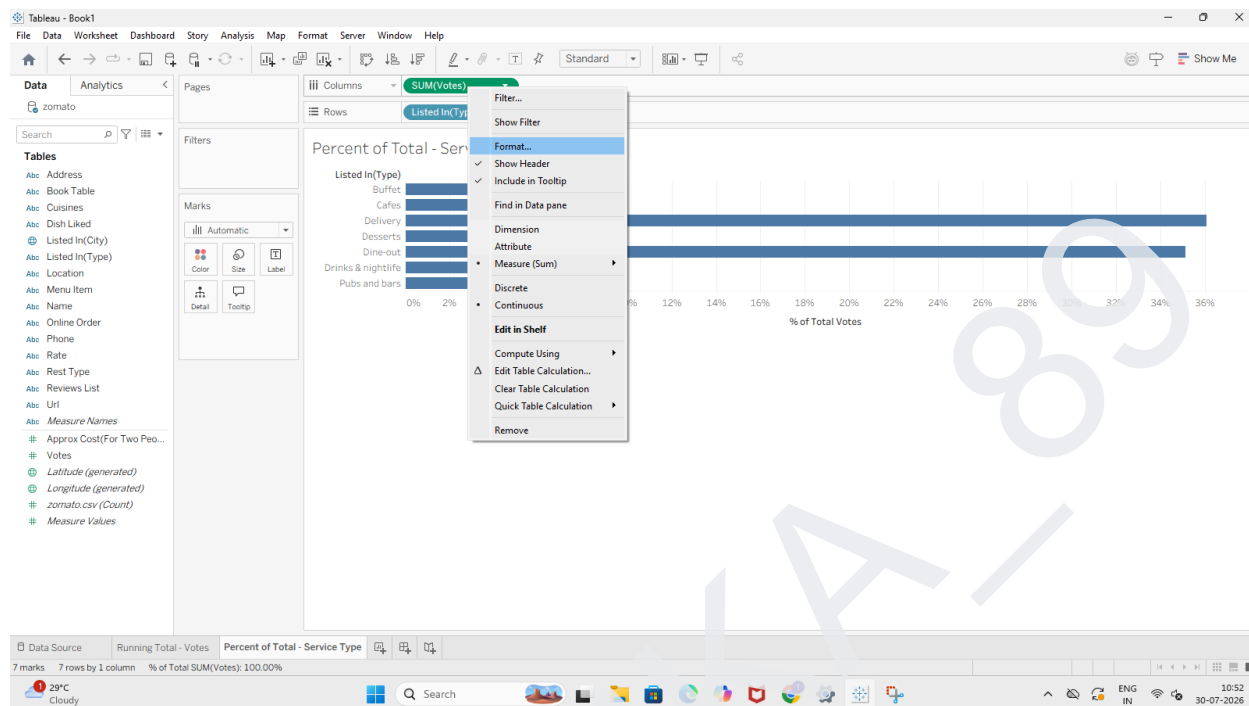
NAME : TANISHKA KUTWAL

ROLL NO.:- T089

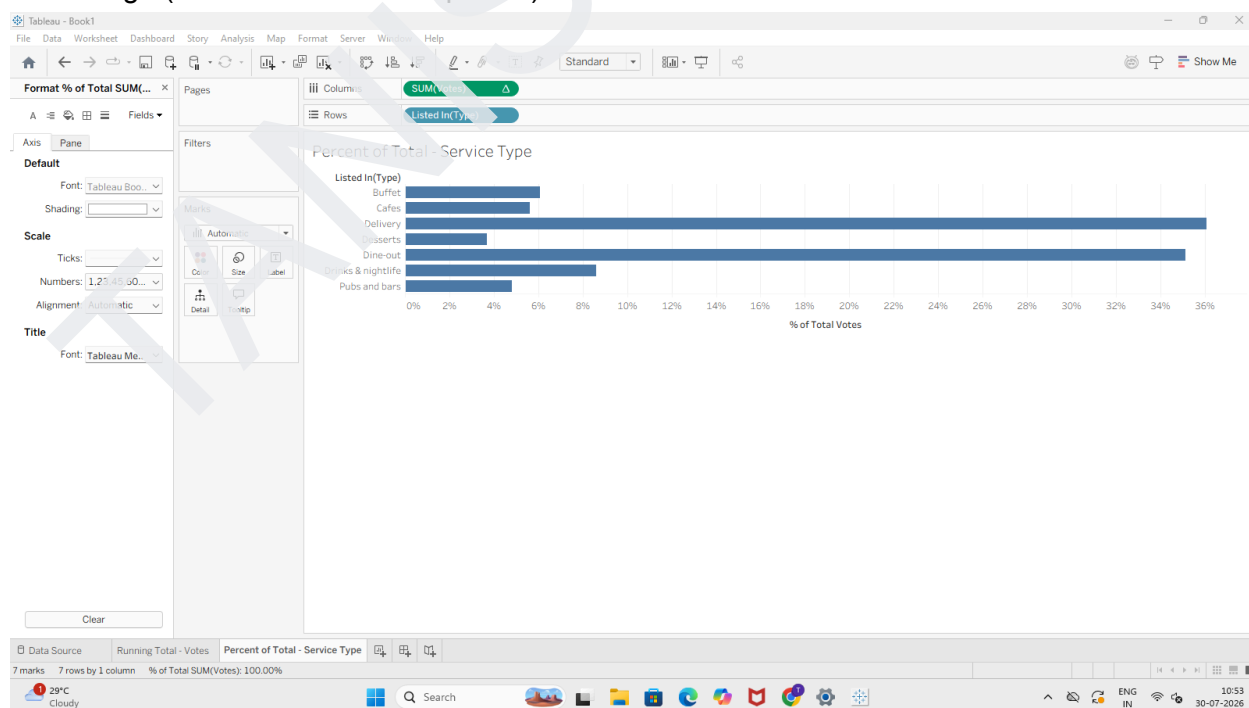
SHETH L.U.J. AND SIR M.V. COLLEGE

SUB:-Data Visualization with Power BI and Tableau

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Under the Pane tab on the left formatting panel, click Numbers $\rightarrow$ Percentage (set to 1 or 2 decimal places).

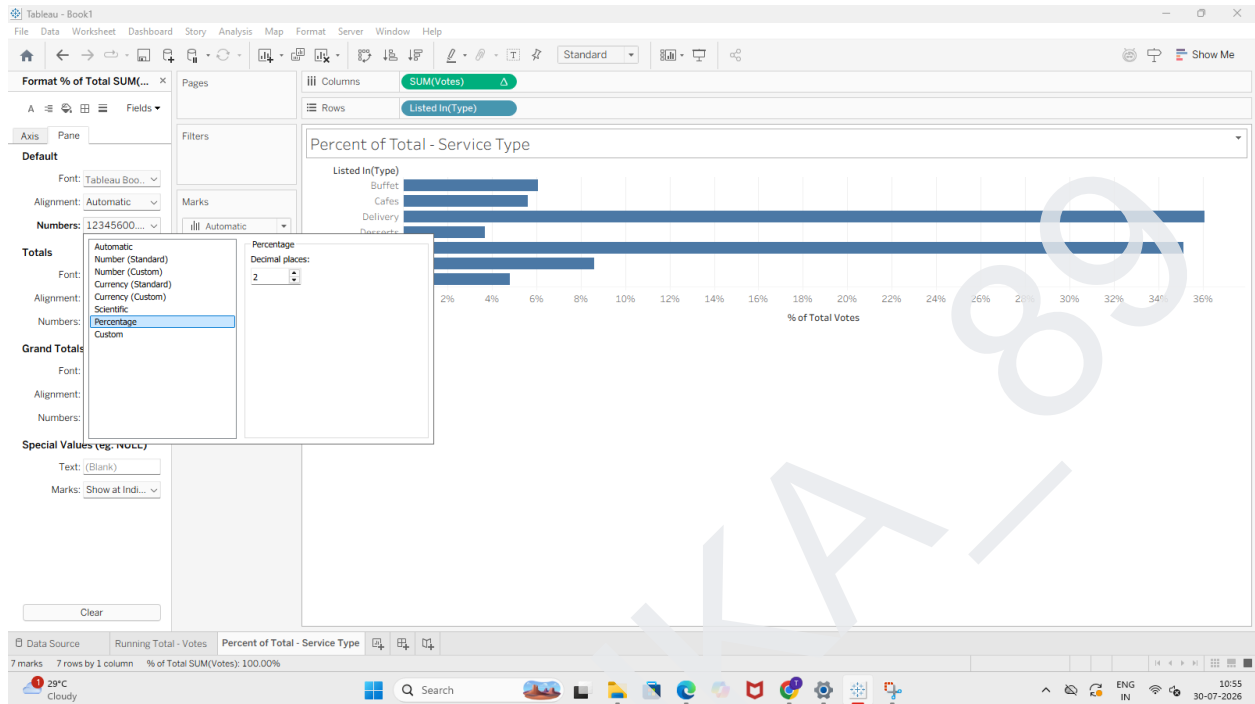


NAME : TANISHKA KUTWAL
ROLL NO:- T089

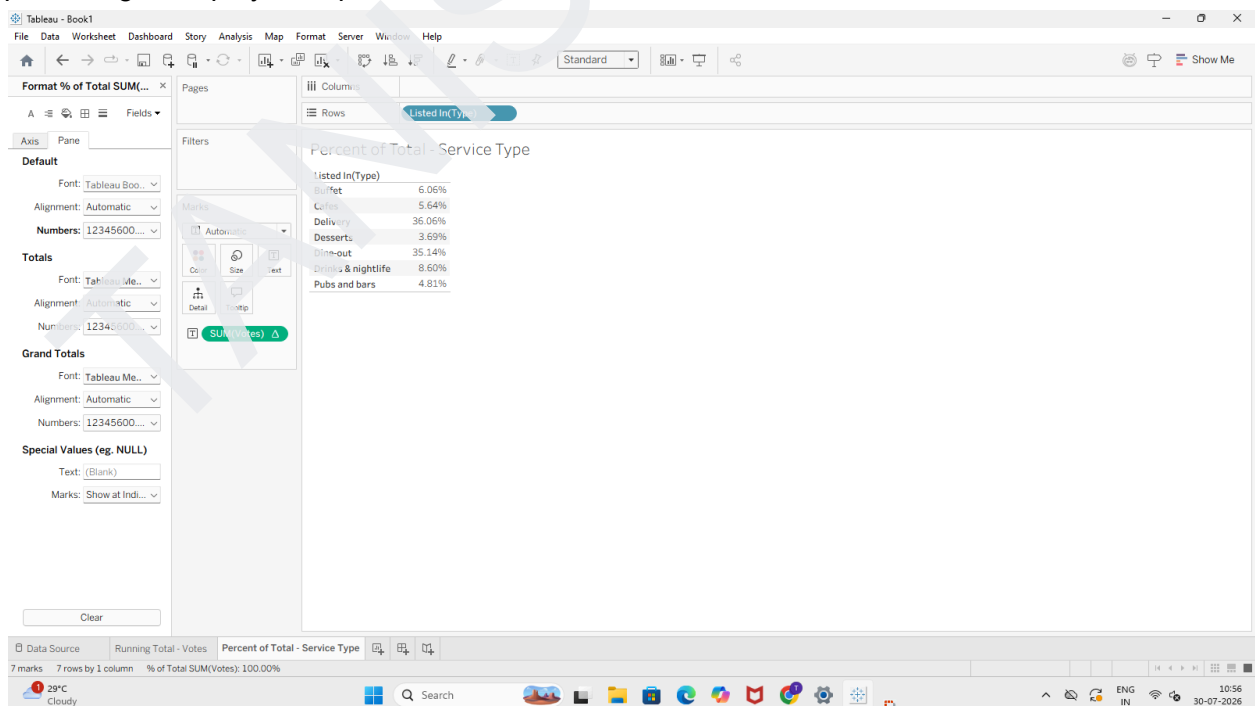
SHETH L.U.J. AND SIR M.V. COLLEGE

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PRACTICAL NO.:-5



Drag SUM(votes) (with the table calculation icon Δ) onto the Label mark so the exact percentages display on top of each bar.



NAME : TANISHKA KUTWAL

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SUB:-Data Visualization with Power BI and Tableau

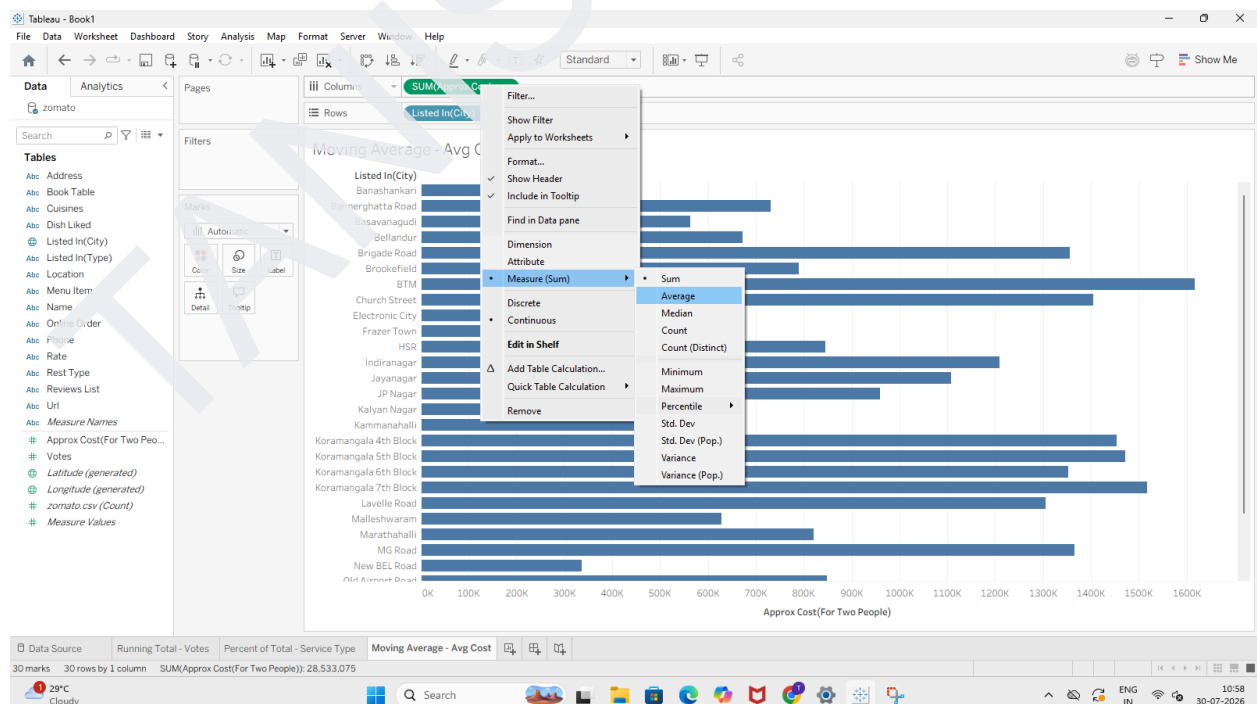
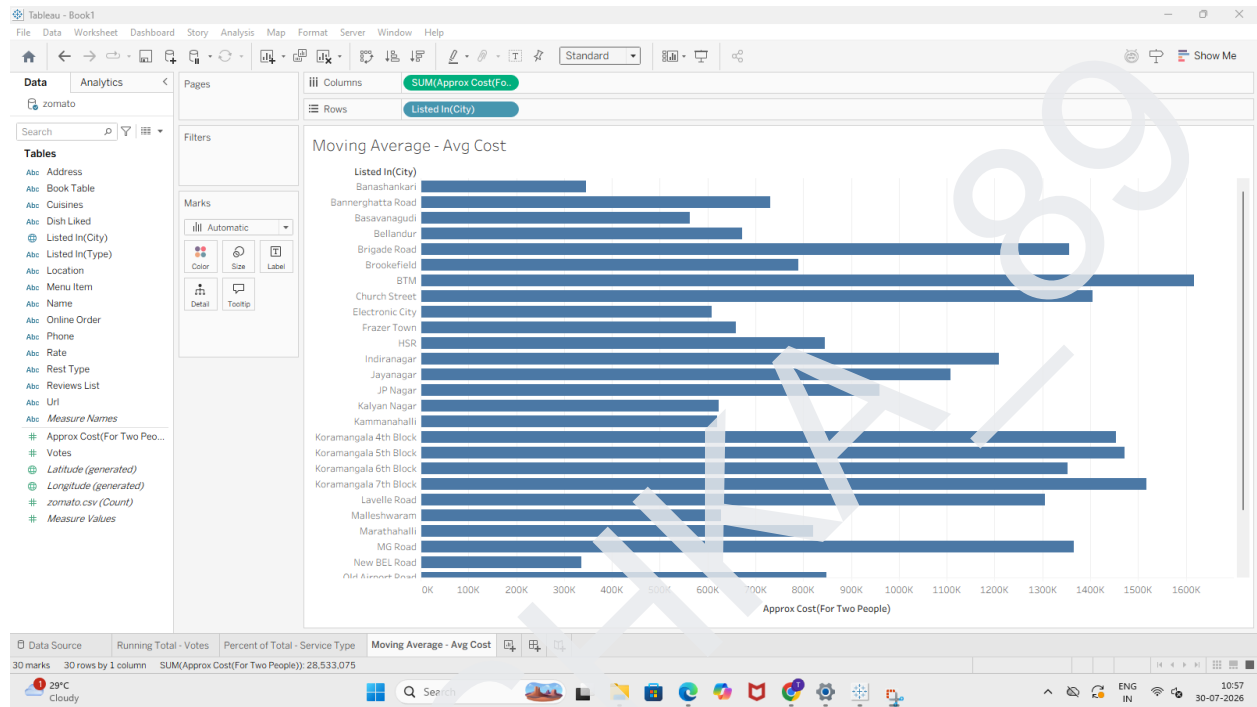
PRACTICAL NO.:-5

B: Create moving averages and cumulative metrics

Open a new worksheet and rename it: Moving Average - Avg Cost.

Drag listed_in(city) to the Rows shelf.

Drag approx_cost(for two people) to the Columns shelf.



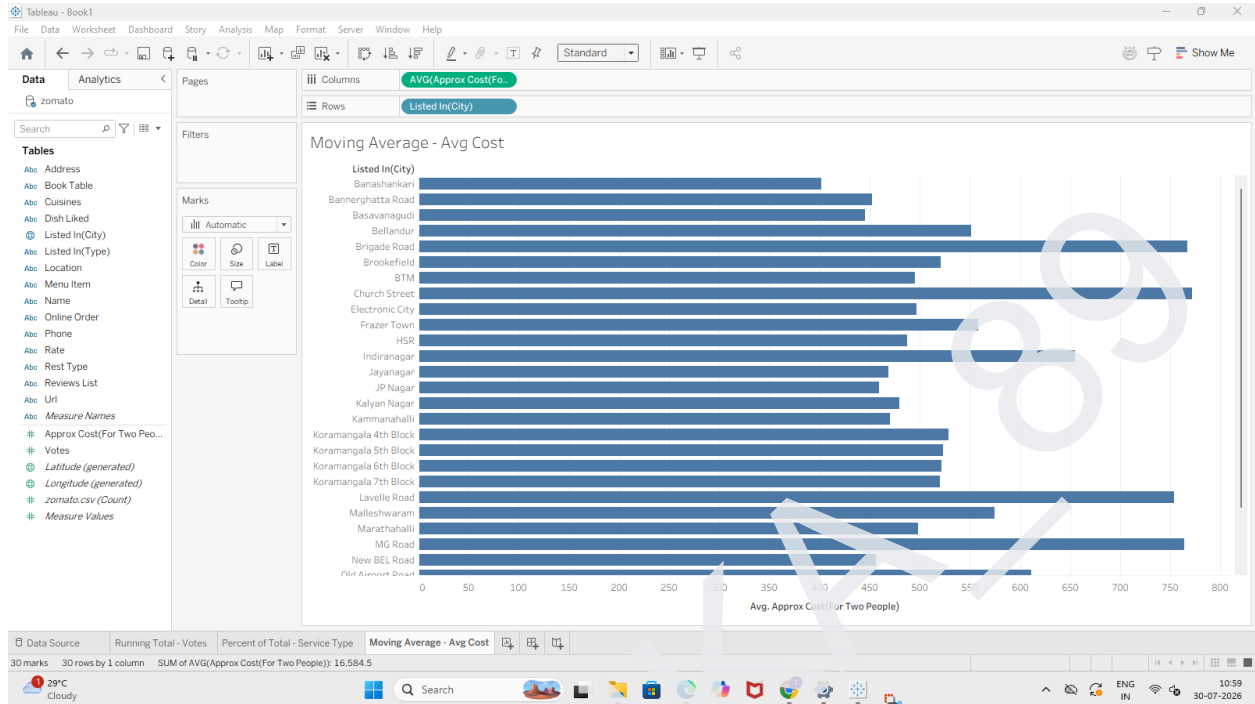
NAME : TANISHKA KUTWAL

ROLL NO:- T089

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SUB:-Data Visualization with Power BI and Tableau

PRACTICAL NO.:-5



Convert to a Moving Average Quick Table Calculation:

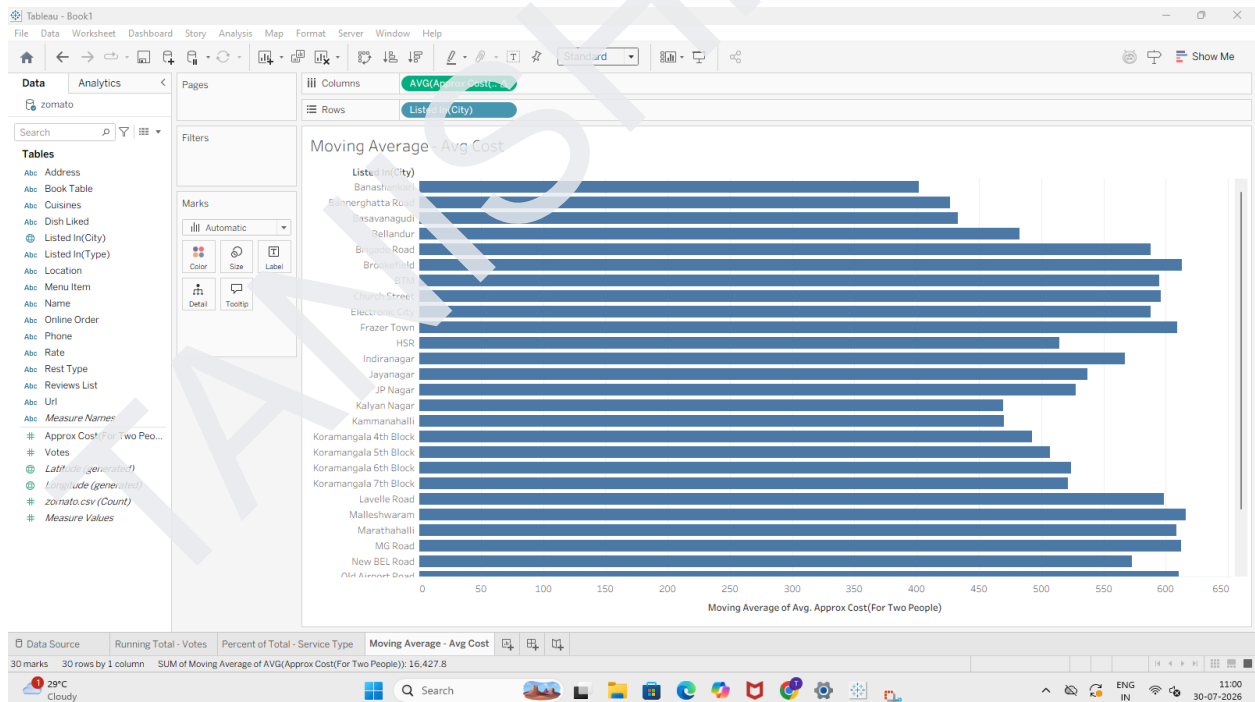
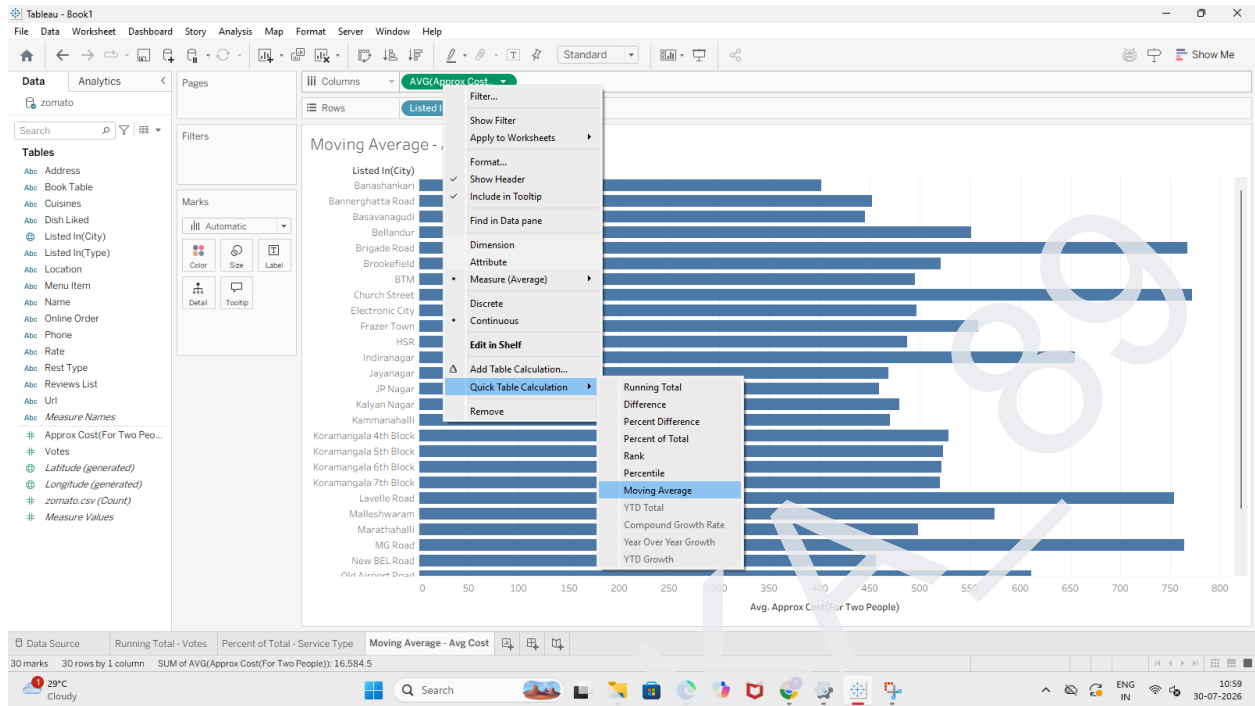
- o Right-click the AVG(approx_cost...) pill on the Columns shelf.
- o Hover over Quick Table Calculation and select Moving Average.

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SUB:-Data Visualization with Power BI and Tableau

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Customize the Moving Window:

- o Right-click the modified AVG(approx_cost...) pill again (it now has a Δ symbol).
- o Select Edit Table Calculation...

NAME : TANISHKA KUTWAL

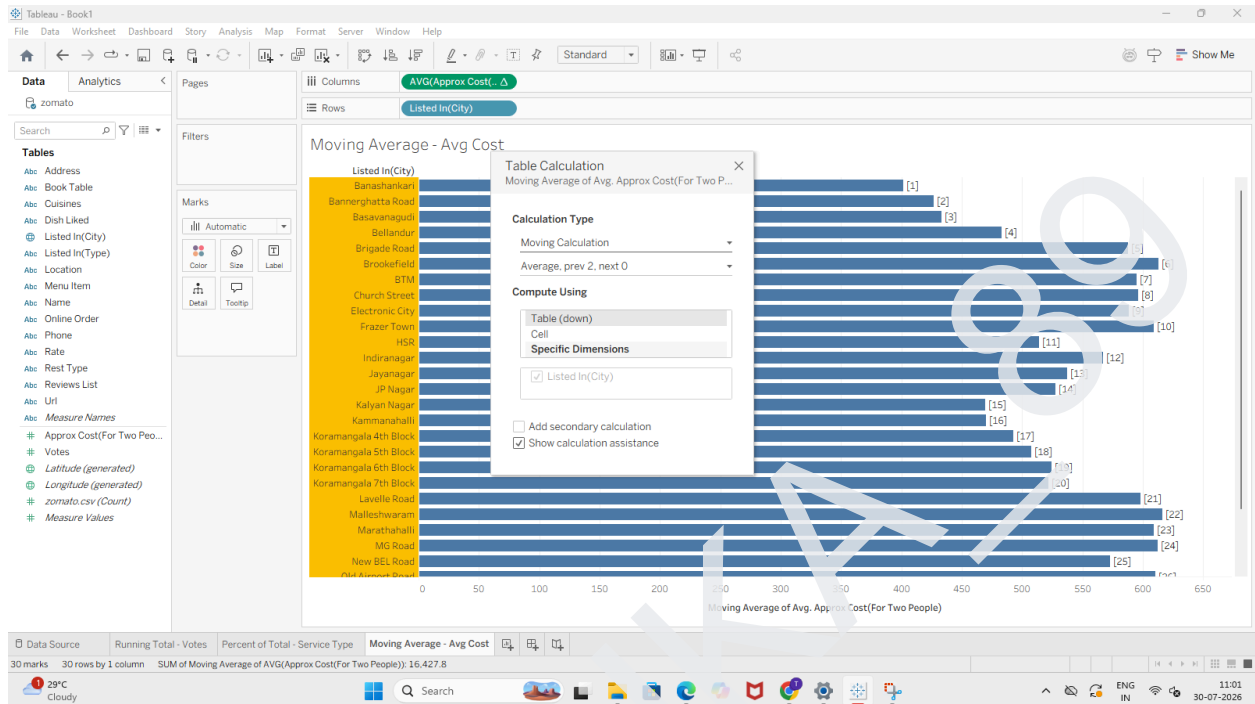
ROLL NO:- T089

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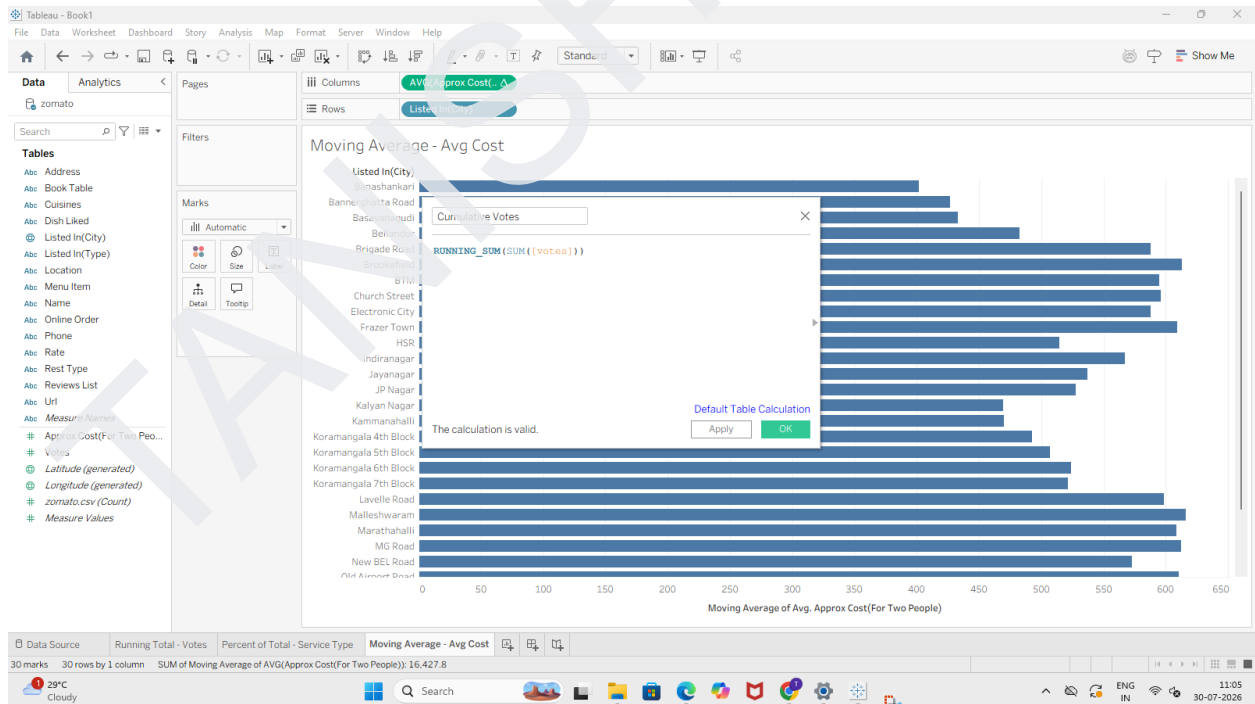
SUB:-Data Visualization with Power BI and Tableau

PRACTICAL NO.-5

In the configuration dialog, set Summarize values using: Average.



Create a Custom Cumulative Metric via Calculated Field



Open a new worksheet.

Drag cuisines to Rows (Filter to top 15 or 20 if the list is too long).

Drag your newly created Cumulative Votes (RUNNING_SUM) field onto Columns.

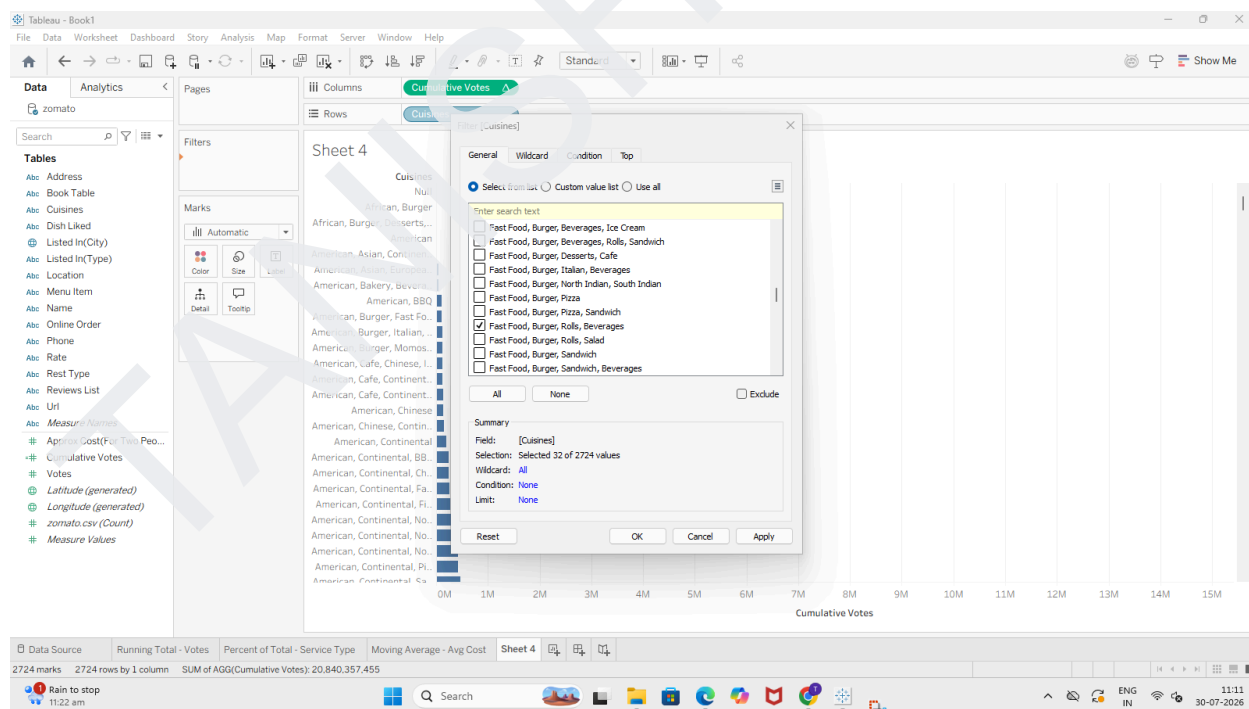
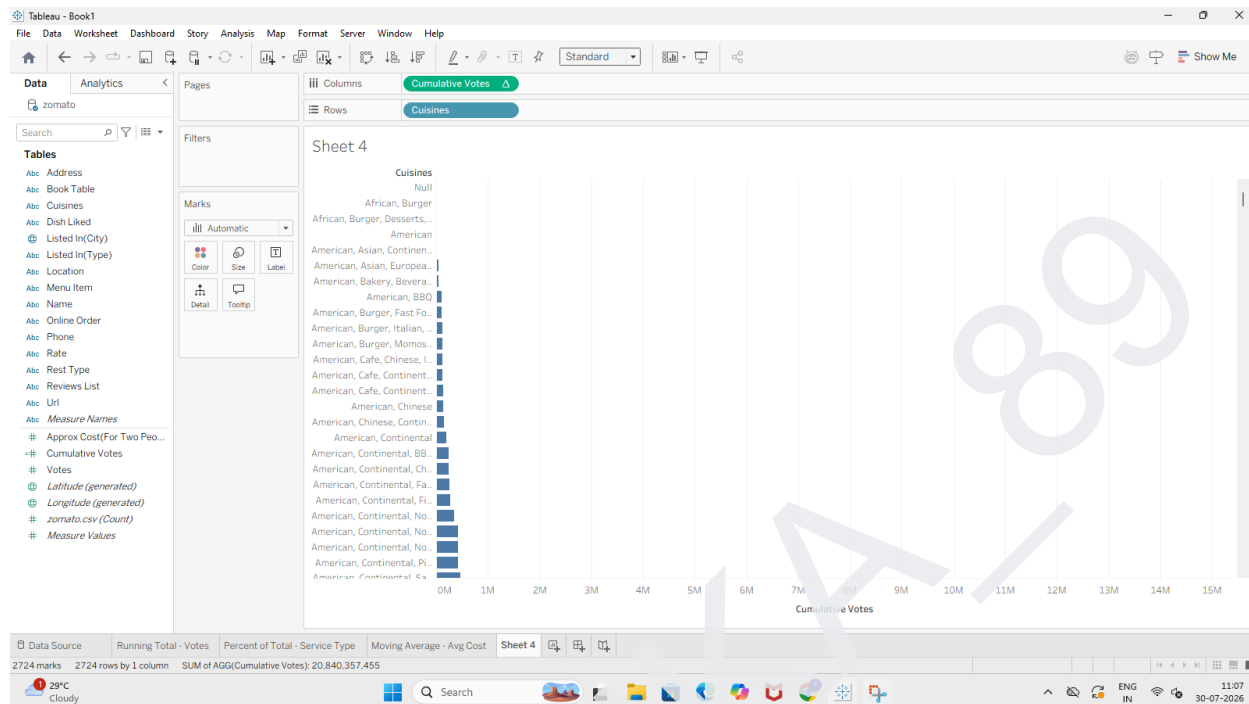
NAME : TANISHKA KUTWAL

ROLL NO:- T089

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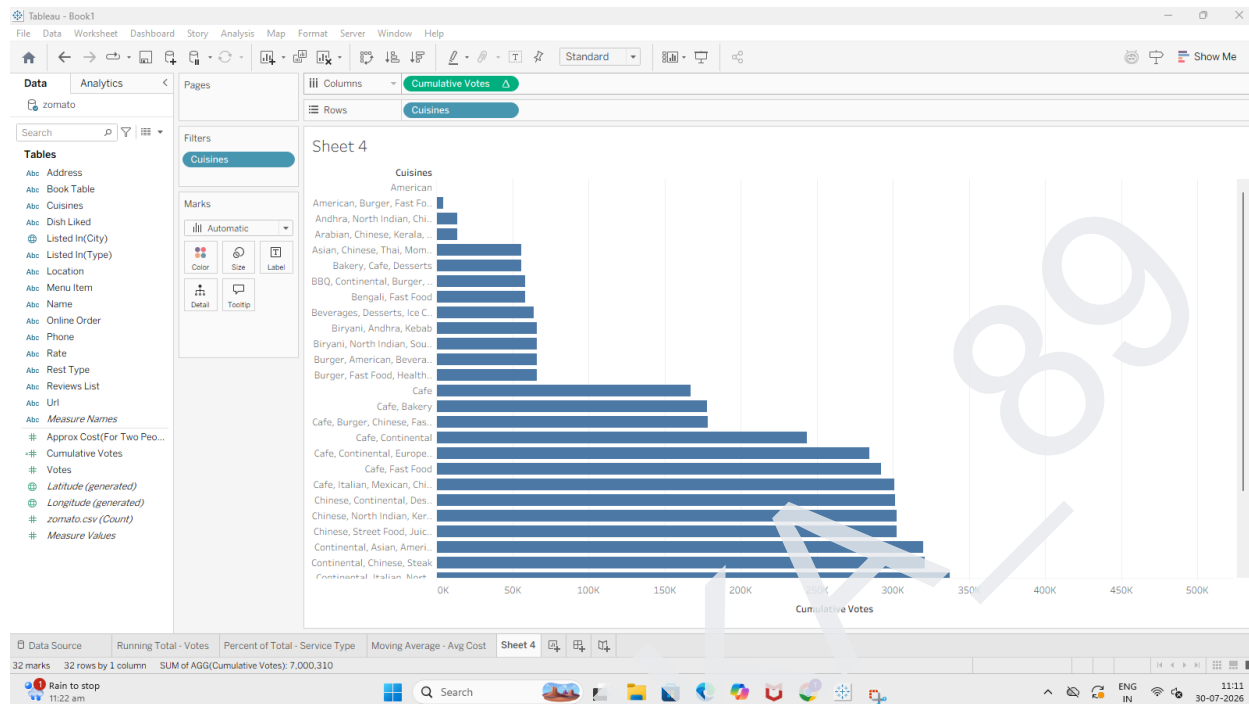


NAME : TANISHKA KUTWAL
ROLL NO:- T089

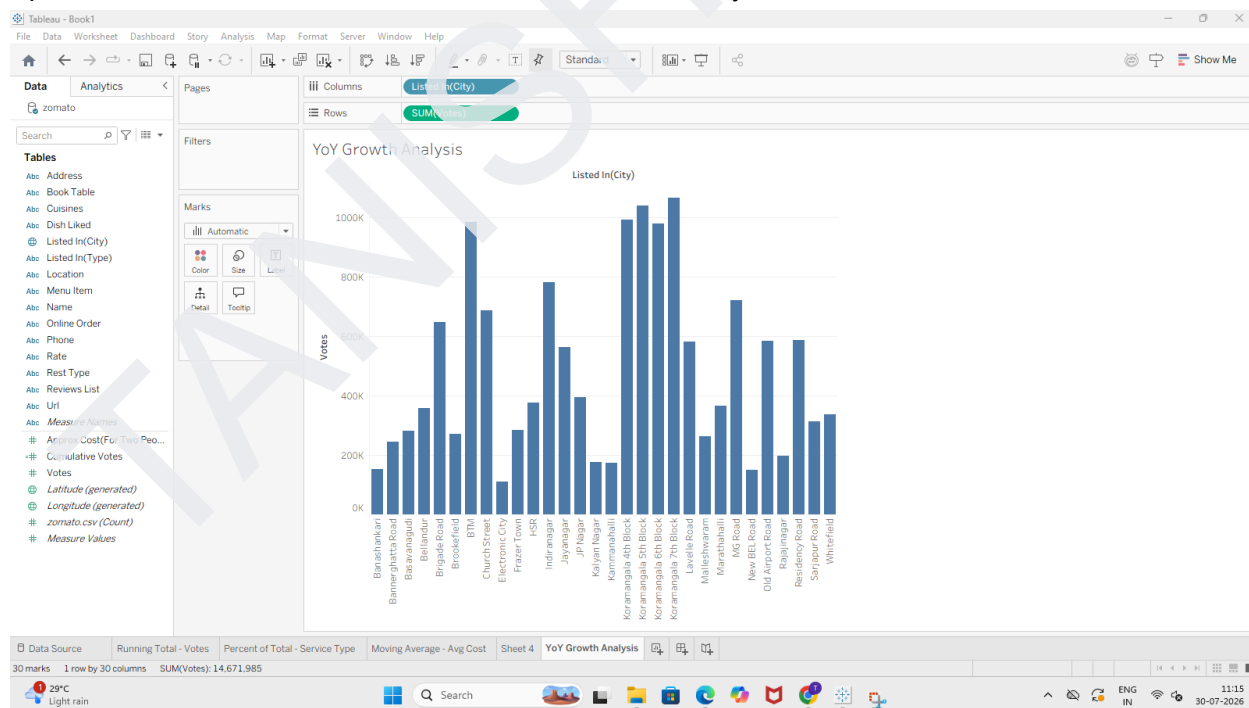
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PRACTICAL NO.:-5



C: Perform year-over-year growth analysis
Open a new worksheet and rename it: YoY Growth Analysis.



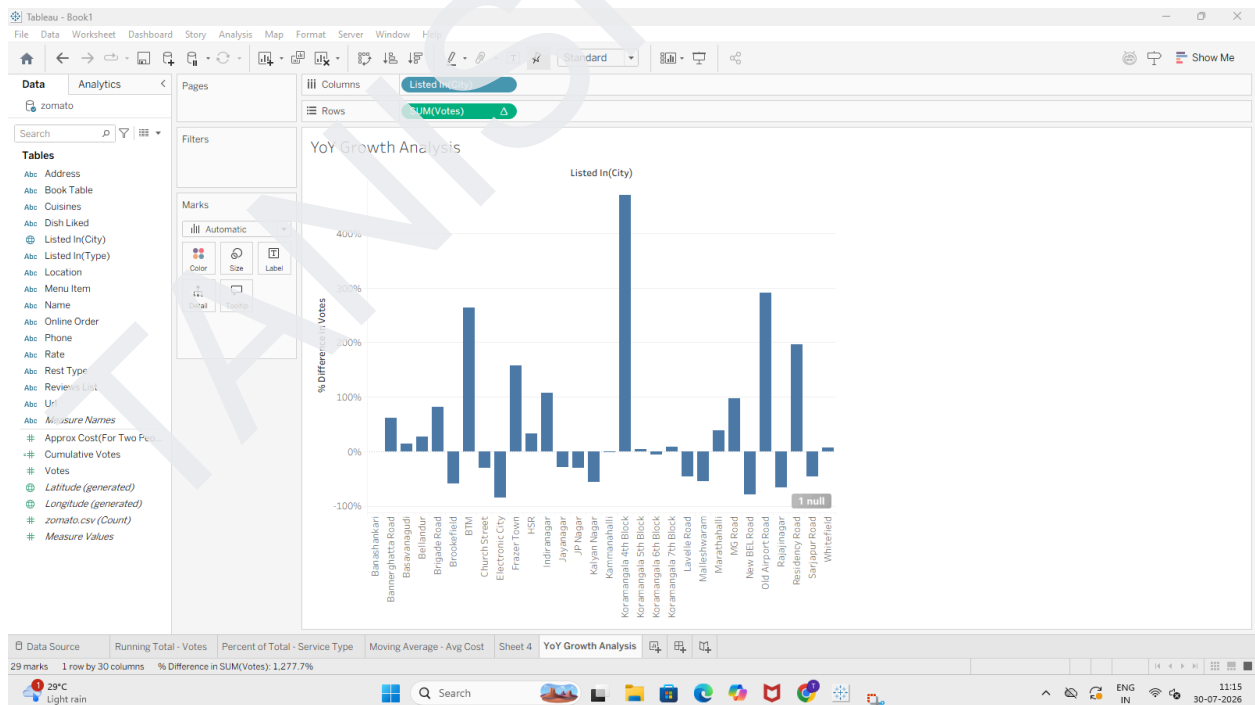
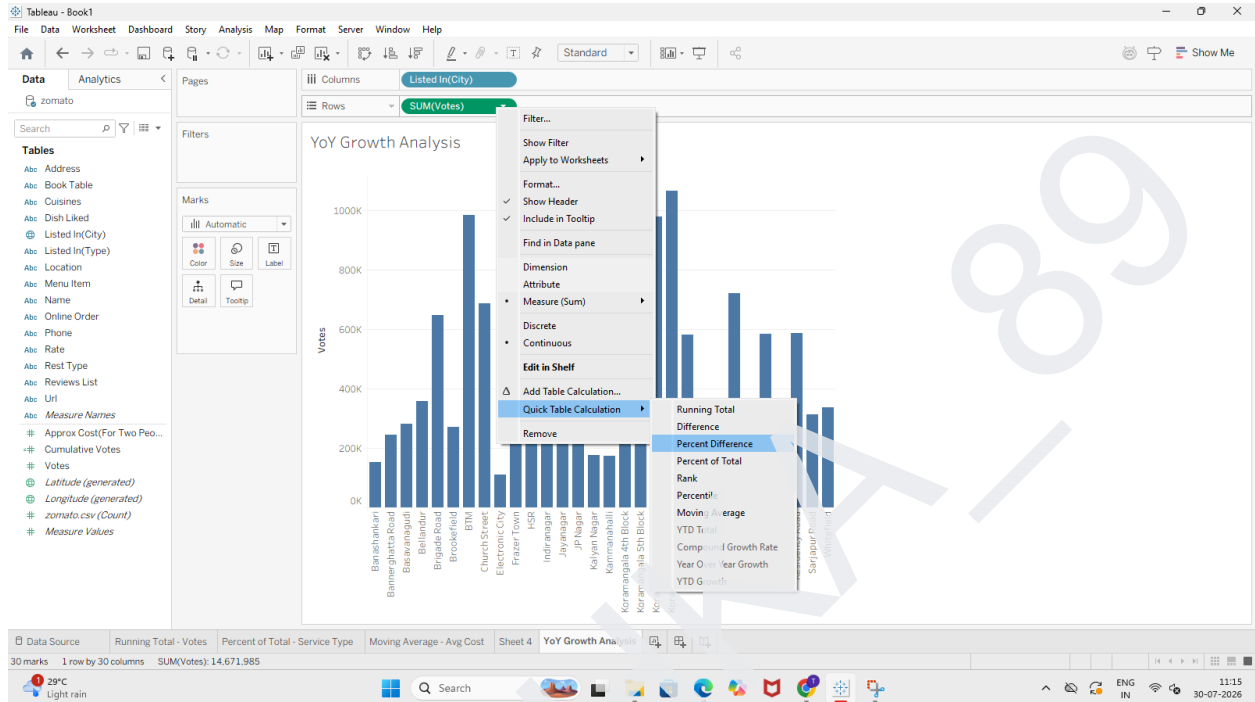
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ROLL NO.:- T089

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Apply the Year-over-Year Growth Quick Table Calculation



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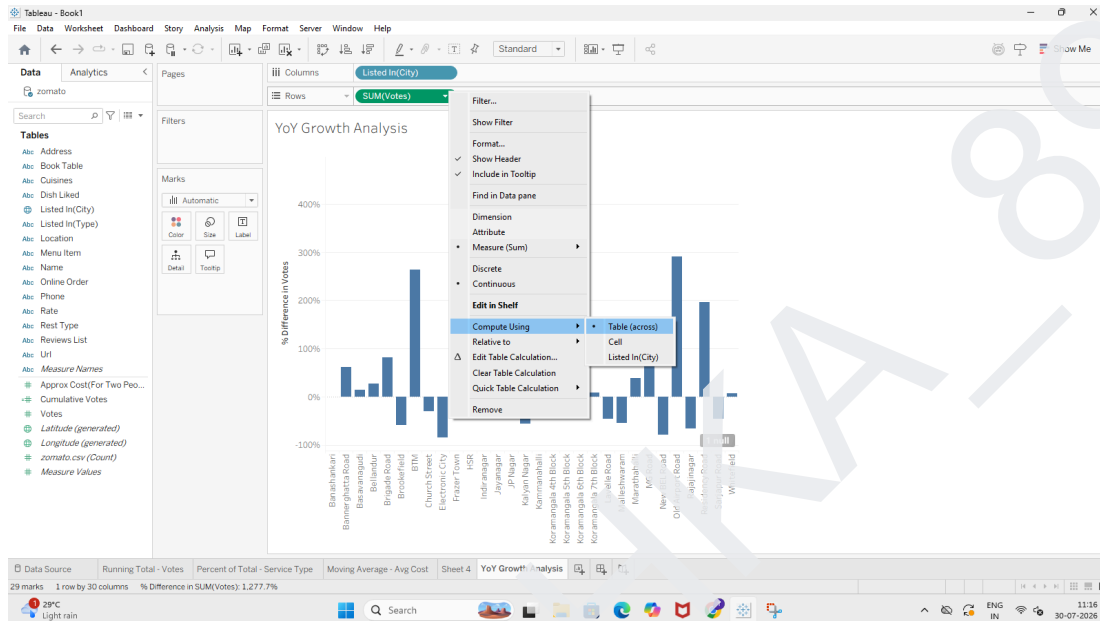
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SUB:-Data Visualization with Power BI and Tableau

PRACTICAL NO.:-5

Configure the Growth Calculation Direction:

- o Right-click the modified SUM(votes) pill (with the Δ icon).
- o Hover over Compute Using select Table (across){In some cases it may have by selected by default}.



Add Visual Color Indicators

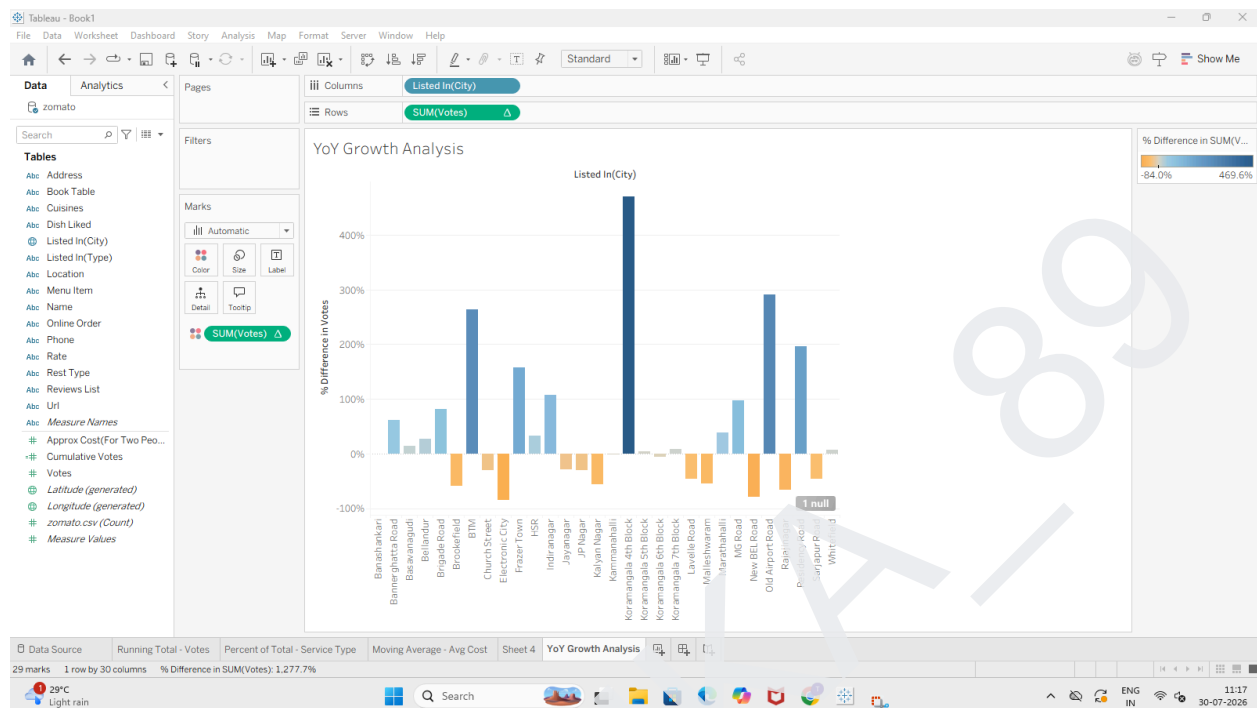
D: Use ranking functions (Top N, Bottom N)

NAME : TANISHKA KUTWAL
ROLL NO:- T089

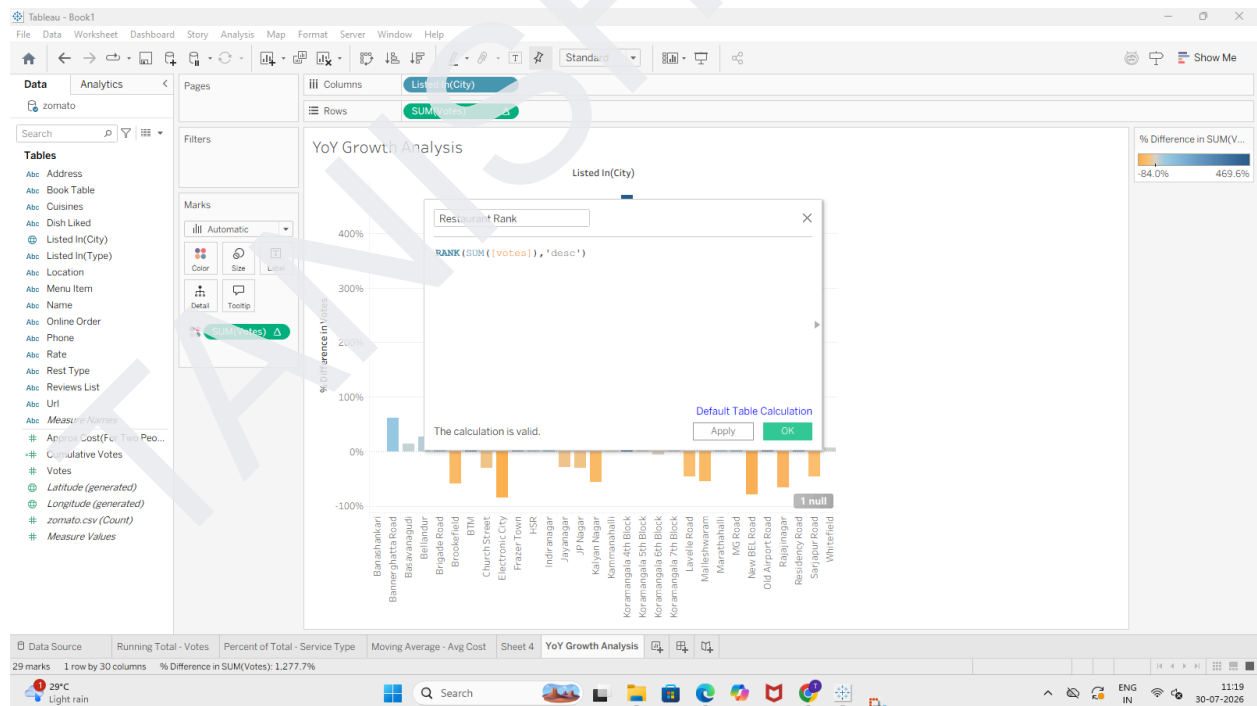
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PRACTICAL NO.:-5



Create a Rank Calculation

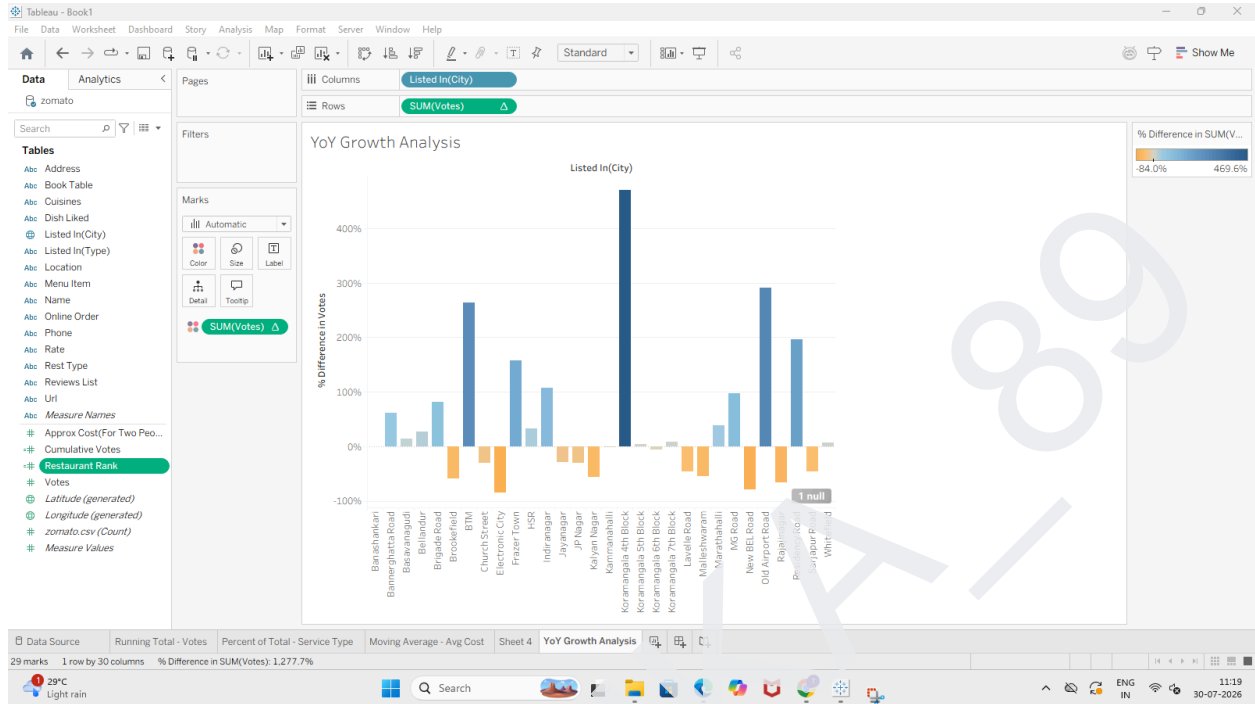


NAME : TANISHKA KUTWAL
ROLL NO:- T089

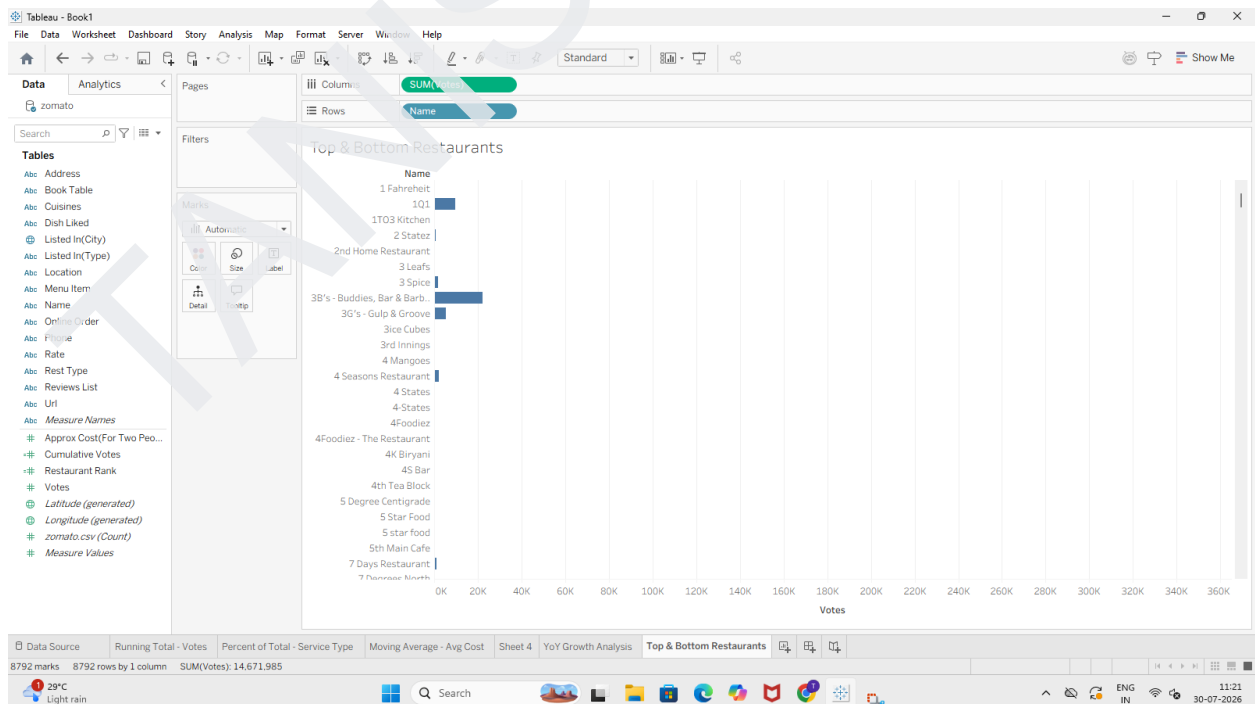
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SUB:-Data Visualization with Power BI and Tableau

PRACTICAL NO.-5



Task 2: Build the Top N / Bottom N Ranked View

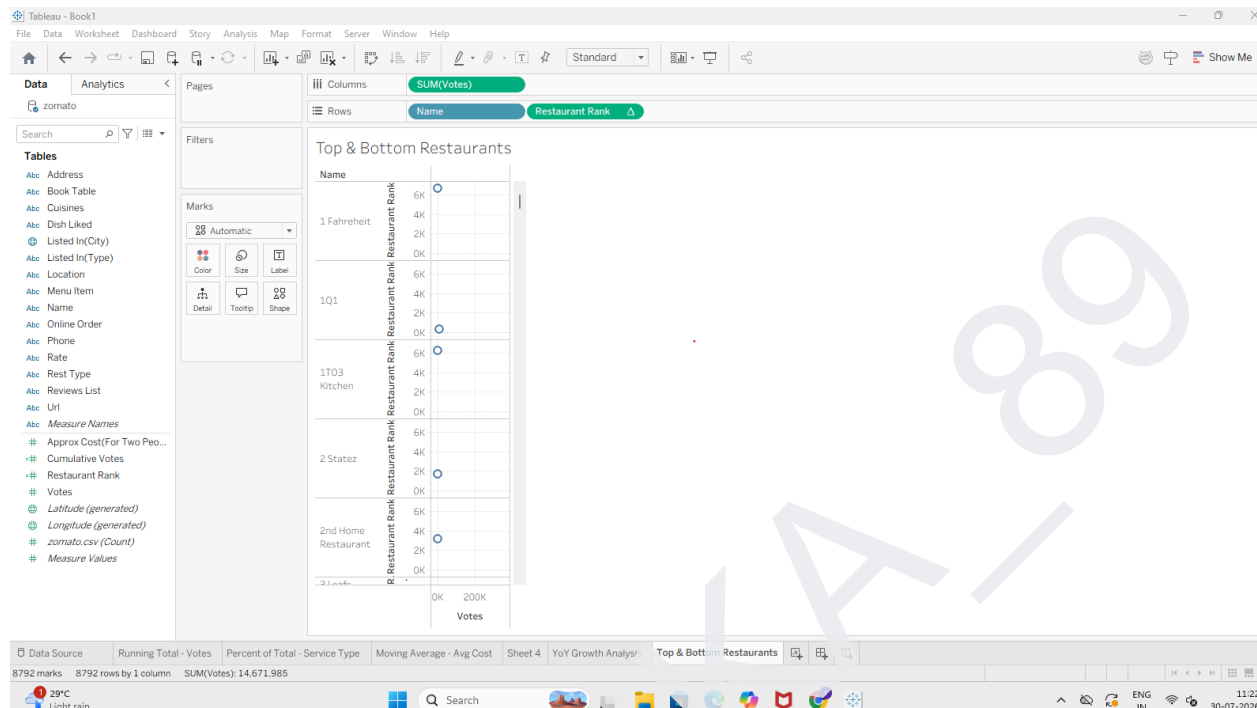


NAME : TANISHKA KUTWAL
ROLL NO:- T089

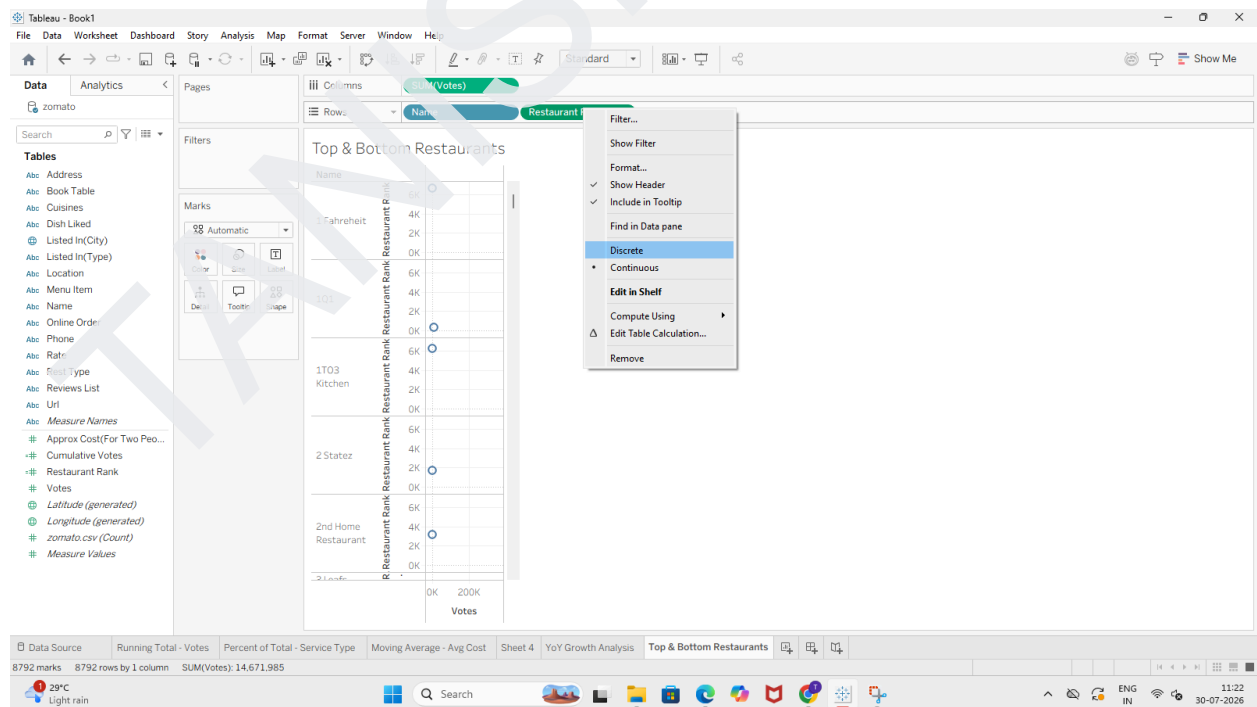
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Convert Restaurant Rank to a Discrete field:

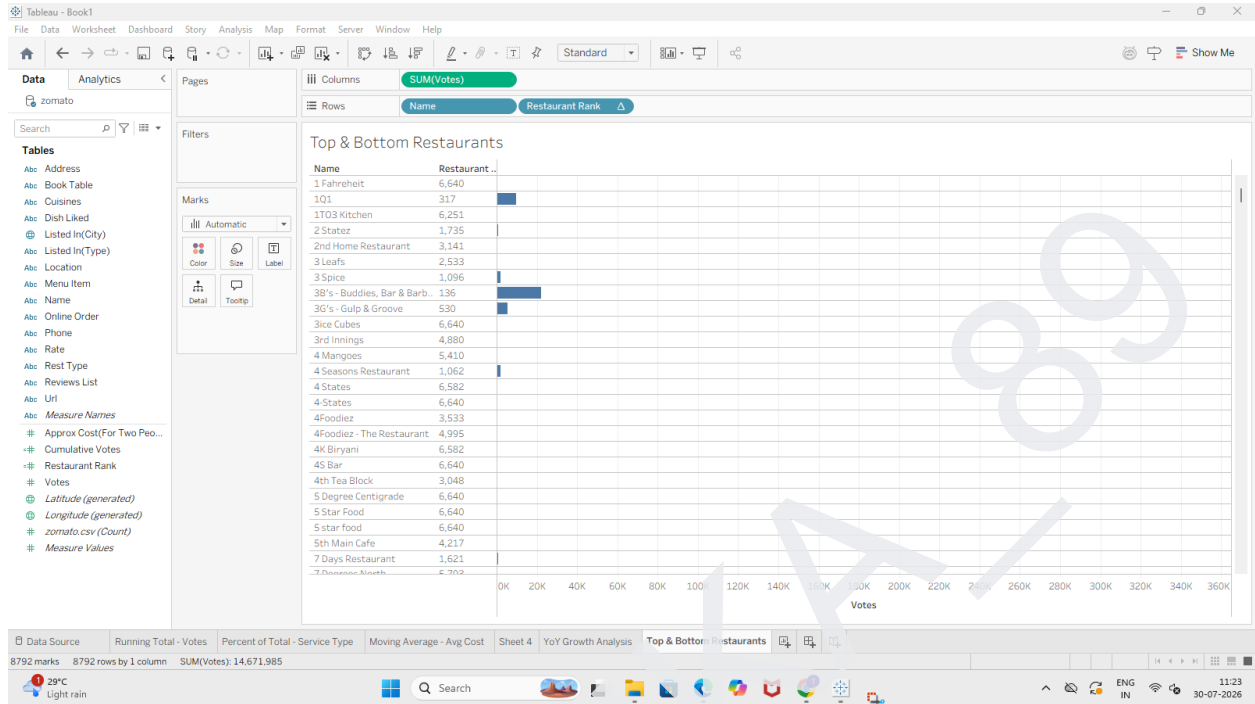


NAME : TANISHKA KUTWAL
ROLL NO:- T089

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SUB:-Data Visualization with Power BI and Tableau

PRACTICAL NO.:-5



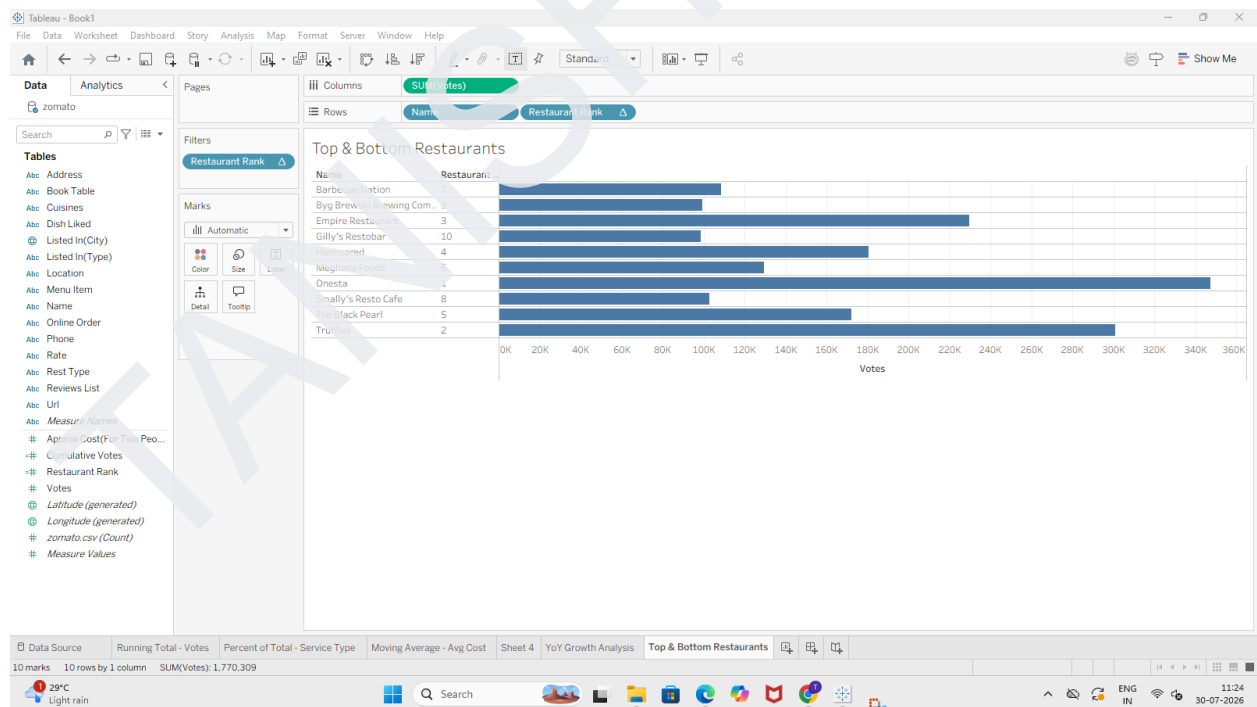
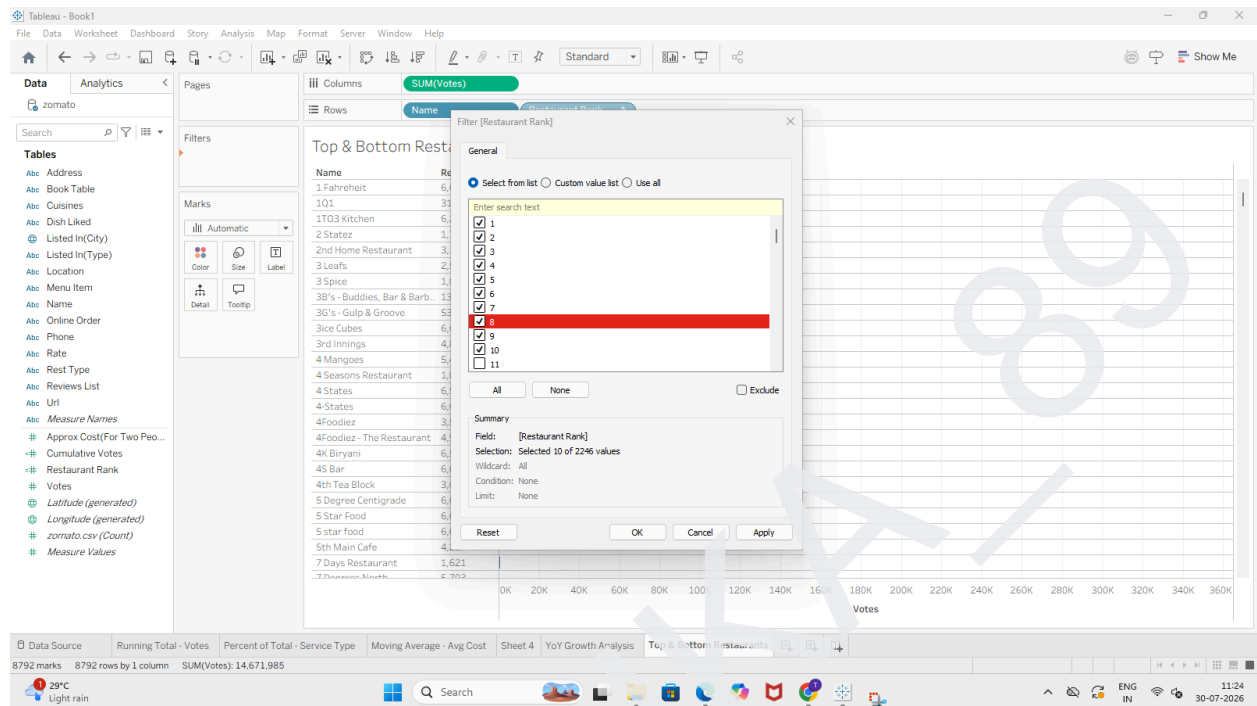
NAME : TANISHKA KUTWAL
ROLL NO:- T089

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SUB:-Data Visualization with Power BI and Tableau

PRACTICAL NO.:-5

Filter Dynamically to Display Top N Only



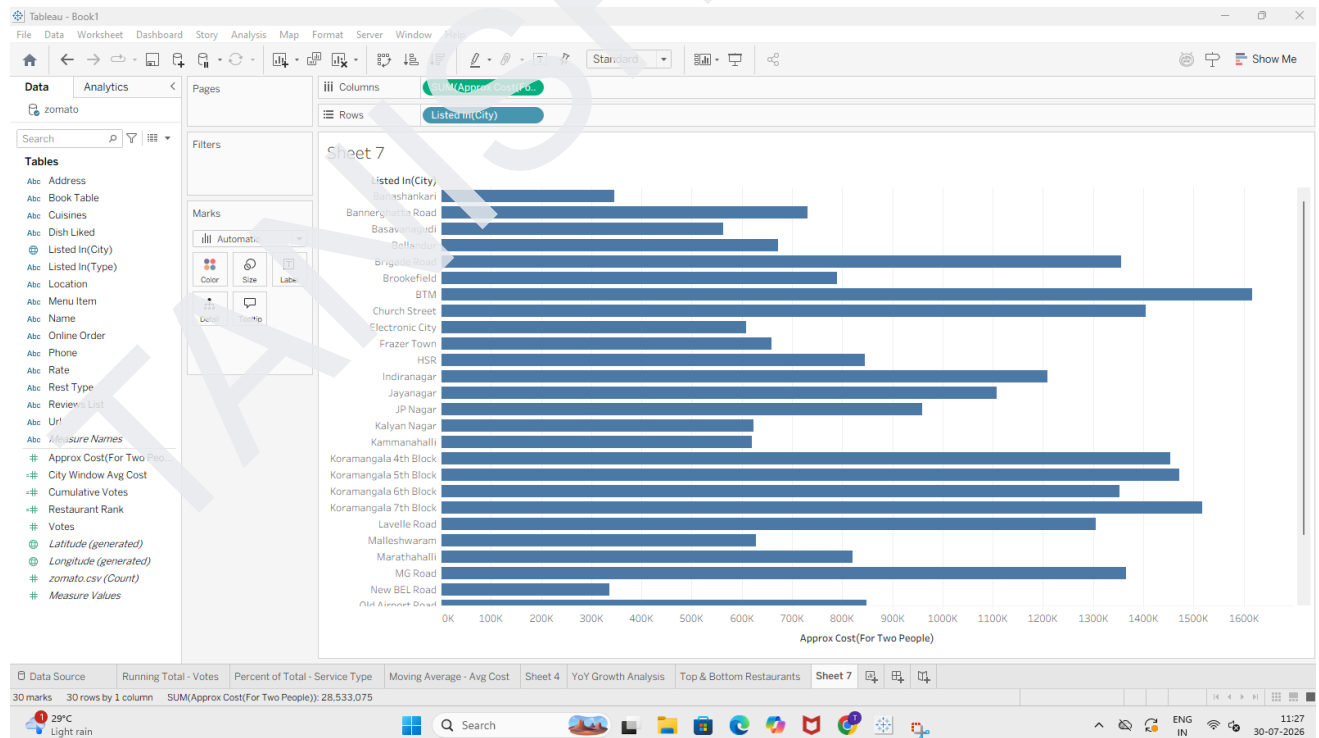
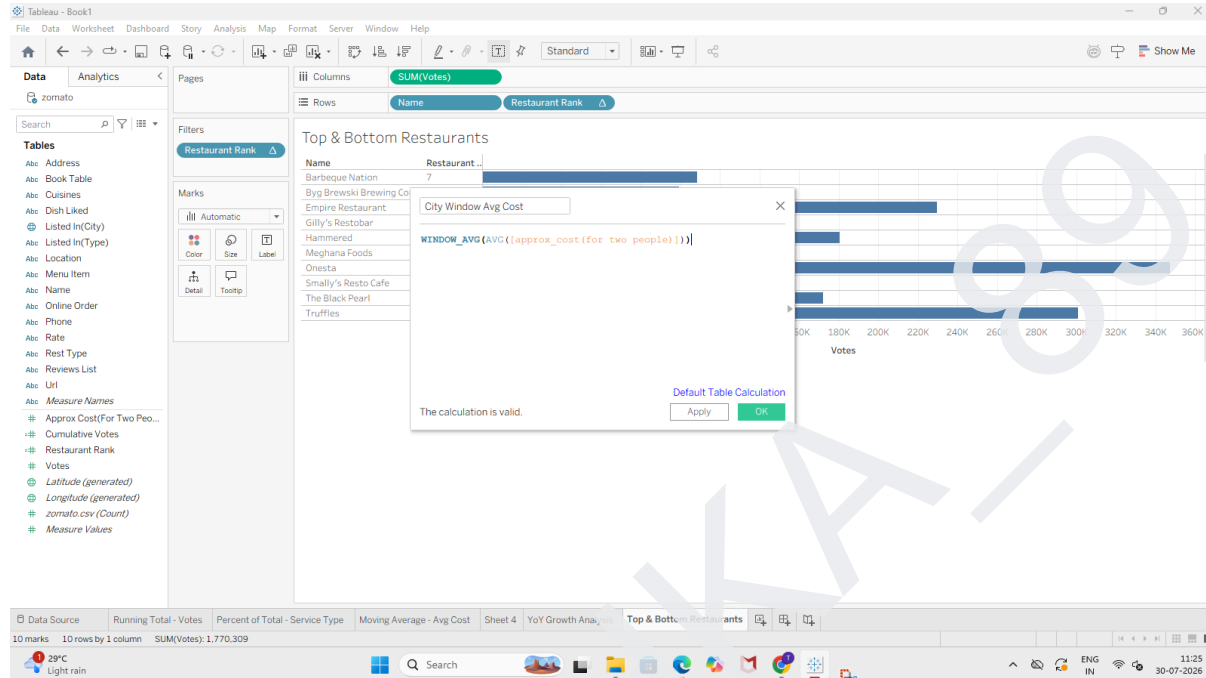
NAME : TANISHKA KUTWAL
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PRACTICAL NO.:-5

E: Implement window functions (WINDOW_SUM, WINDOW_AVG)
Create a Window Average Calculation

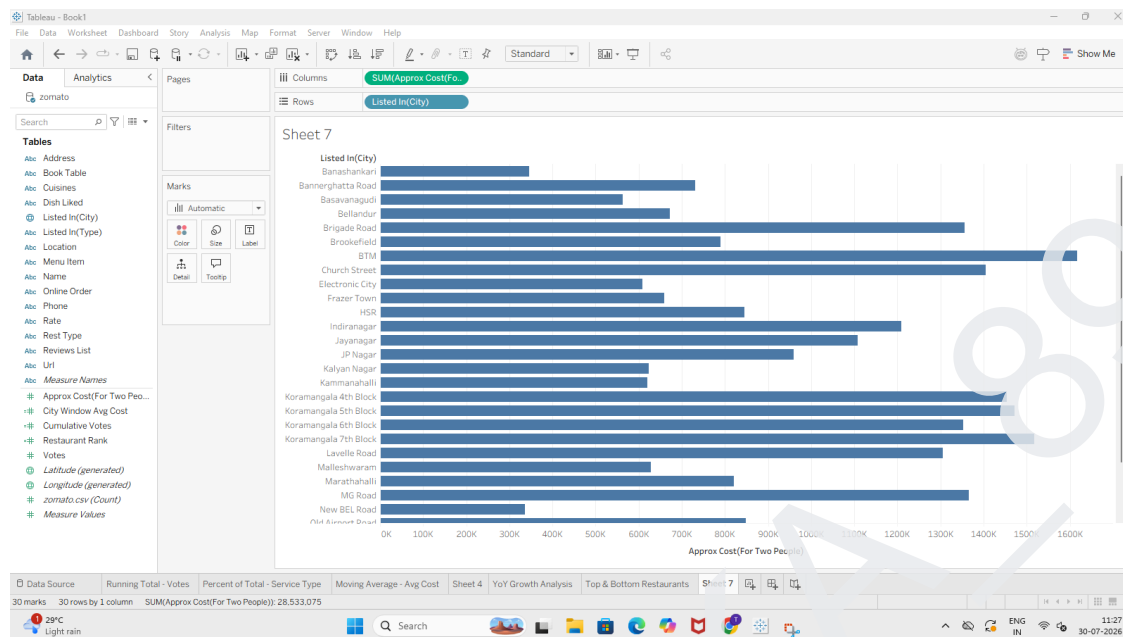


NAME : TANISHKA KUTWAL
ROLL NO:- T089

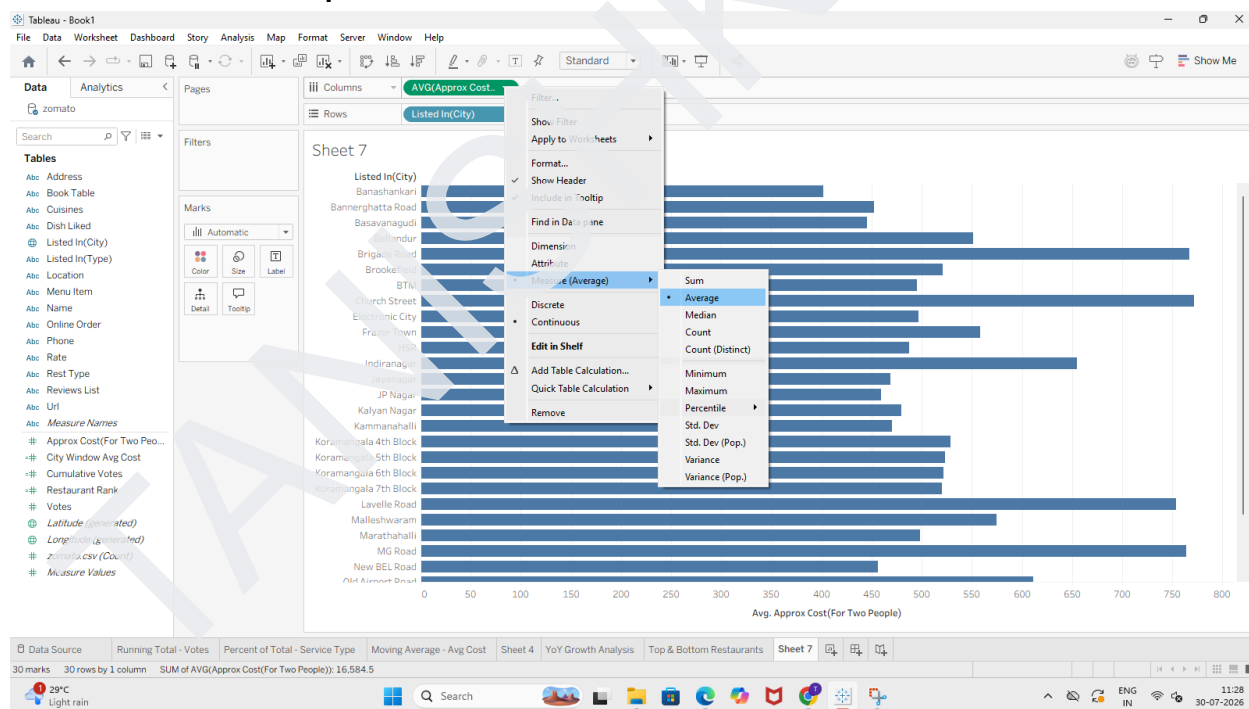
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PRACTICAL NO.-:5



Build the Window Comparison Visualization

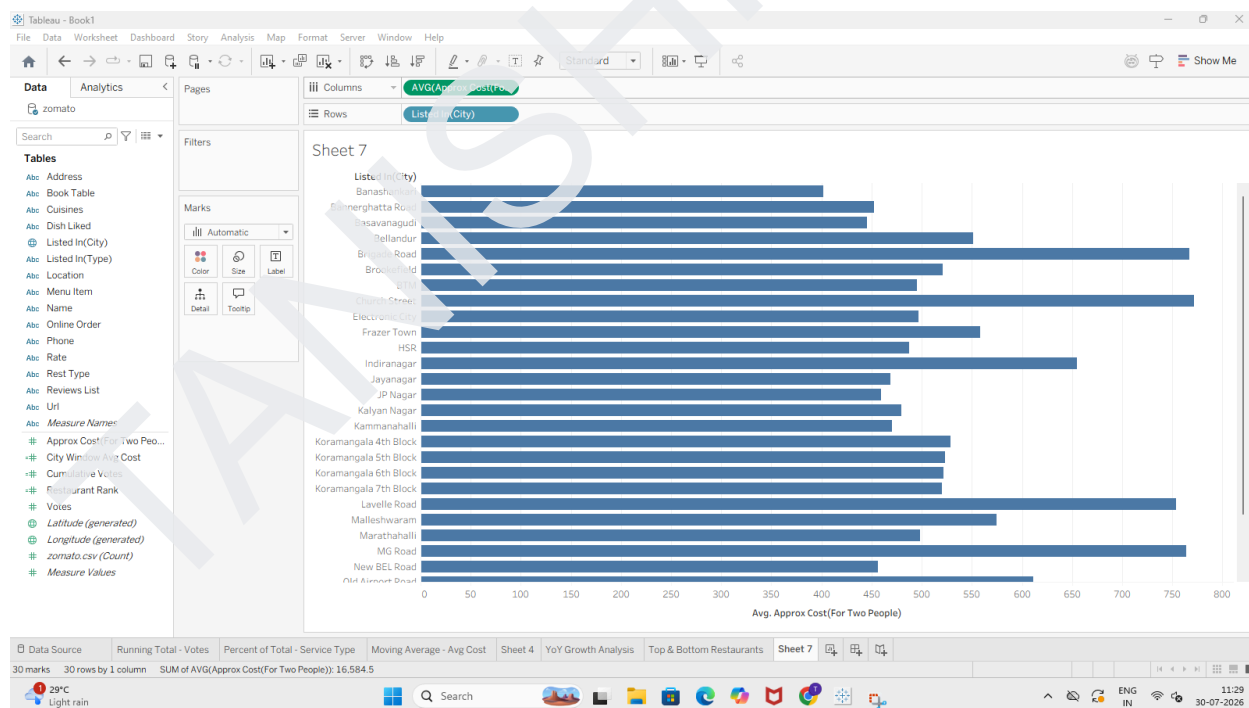
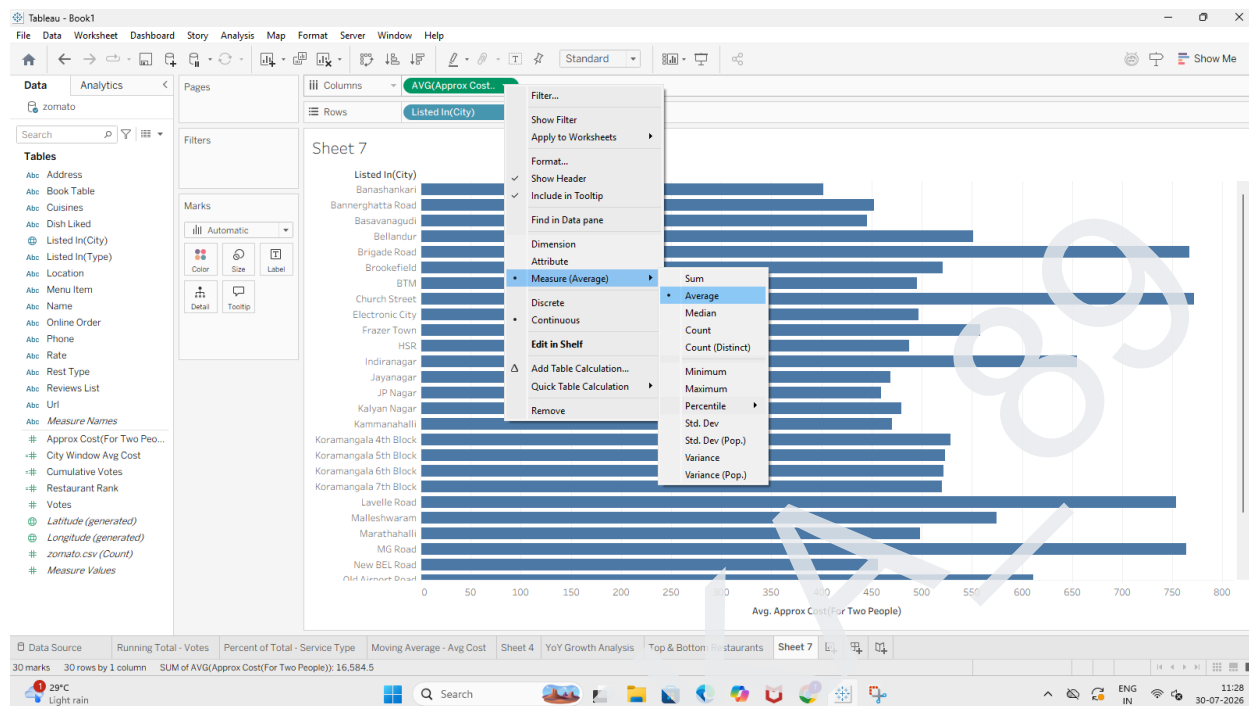


NAME : TANISHKA KUTWAL
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Drag your newly created City Window Avg Cost calculated field onto the Columns

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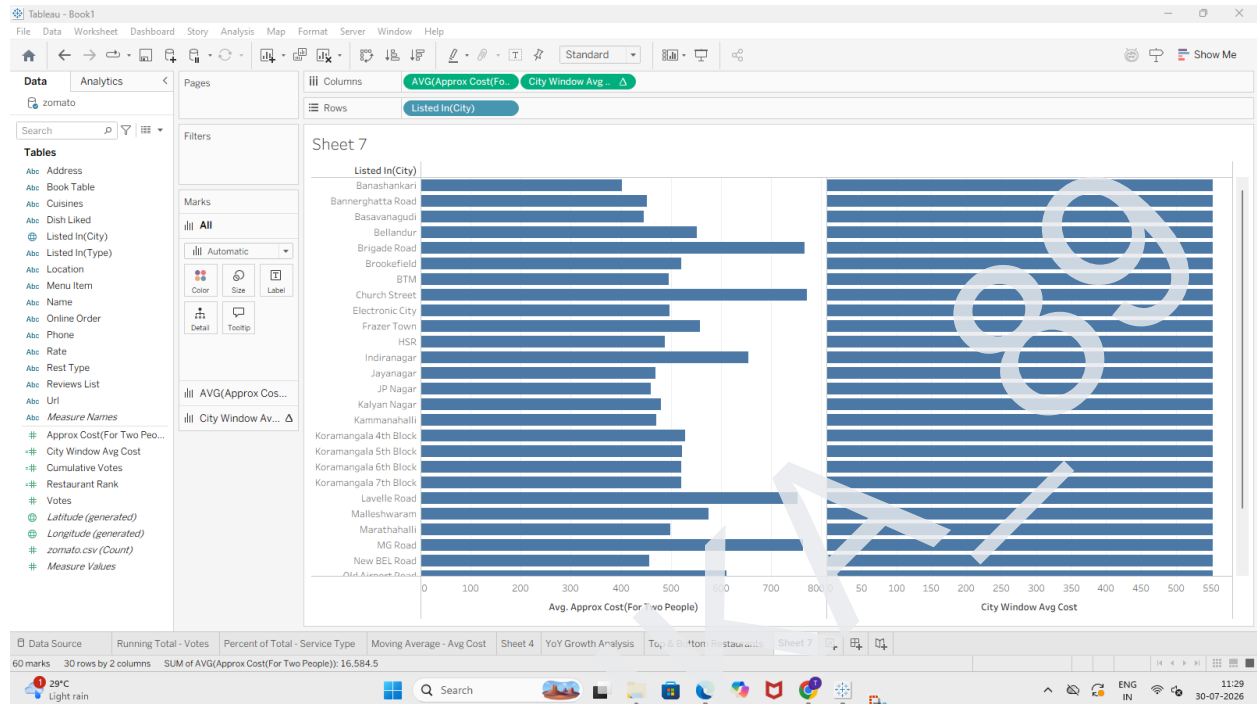
ROLL NO:- T089

SHETH L.U.J. AND SIR M.V. COLLEGE

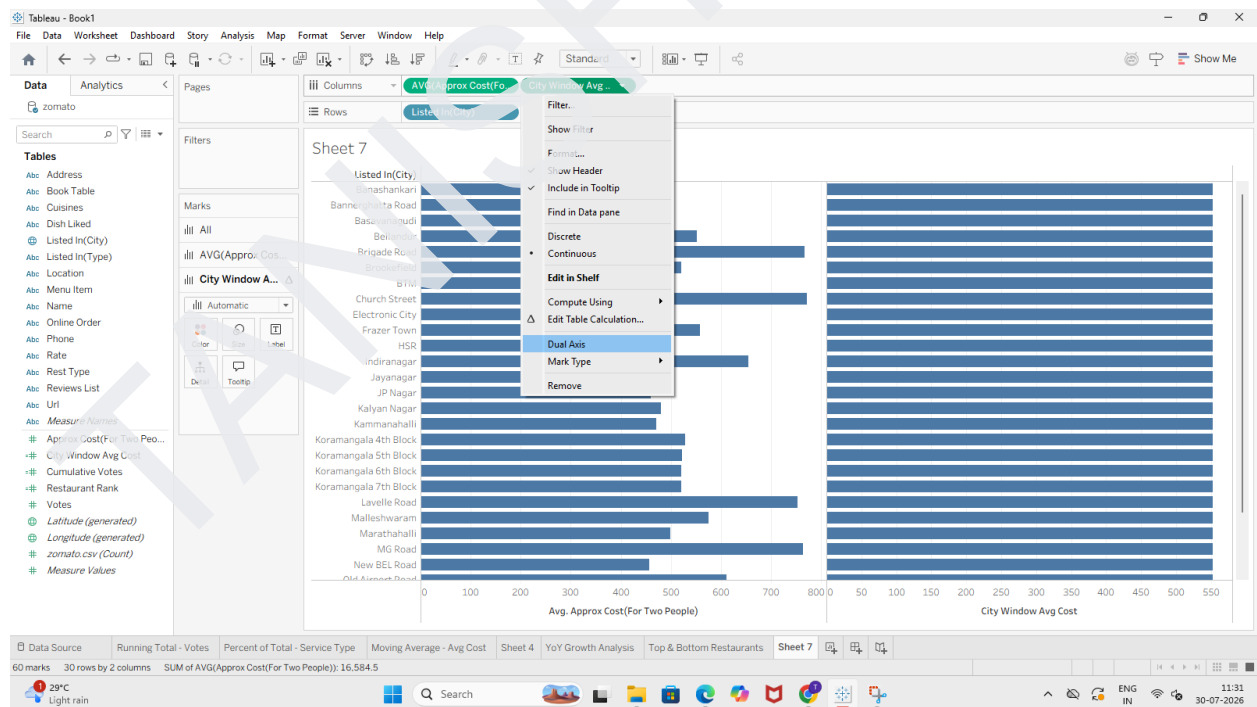
SUB:-Data Visualization with Power BI and Tableau

PRACTICAL NO.:-5

shelf (placing it right beside AVG(approx_cost...)).



Combine Measures into a Dual-Axis Reference View



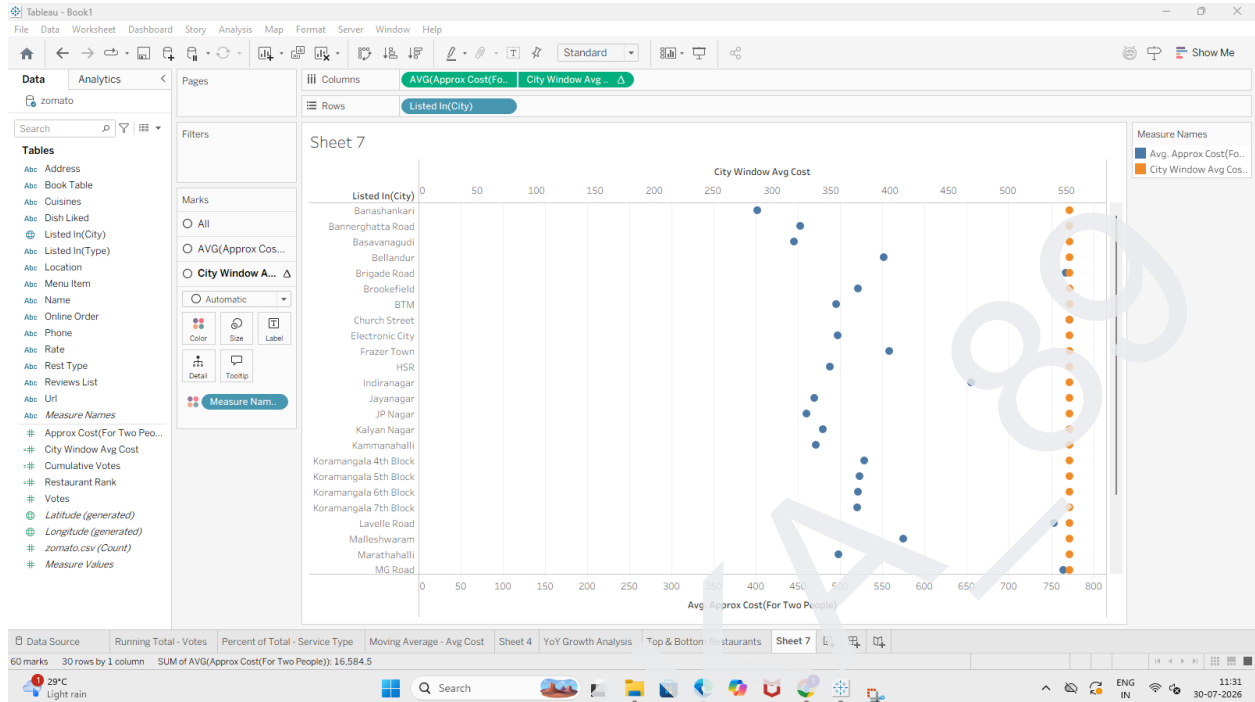
Right-click the top horizontal axis in the visualization view and select Synchronize Axis.

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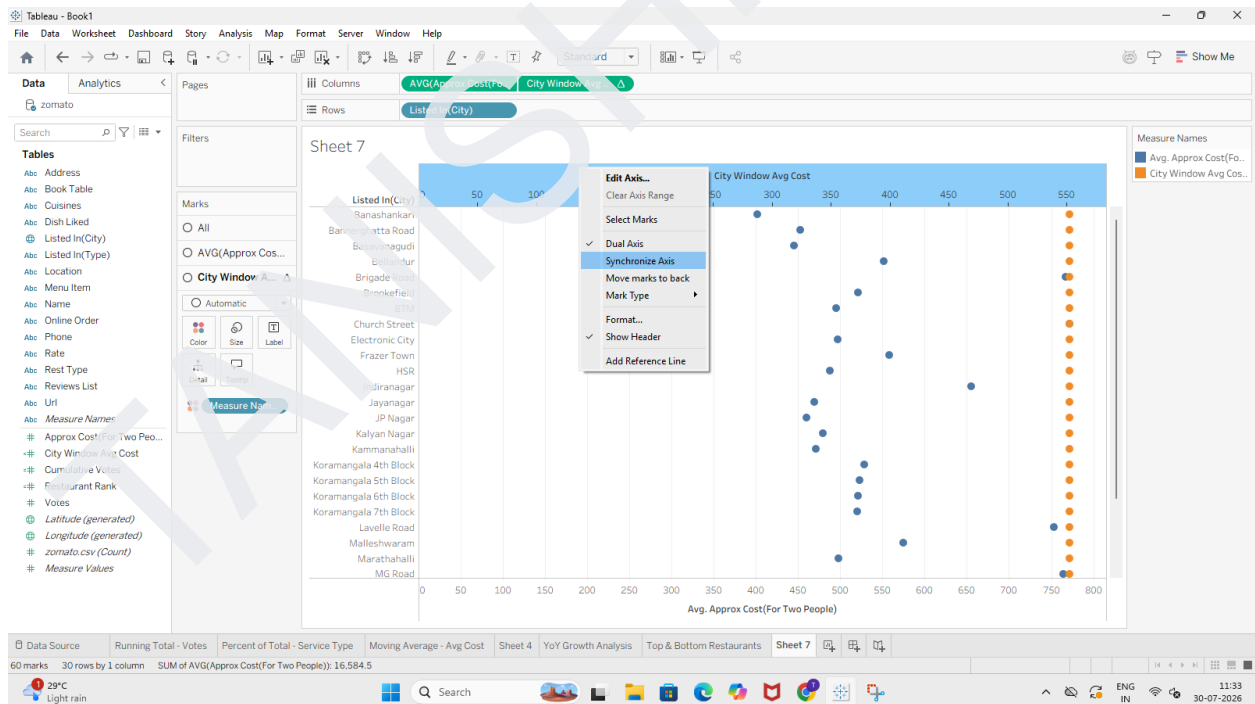
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Change the mark type for AVG(approx_cost...) to Bar.



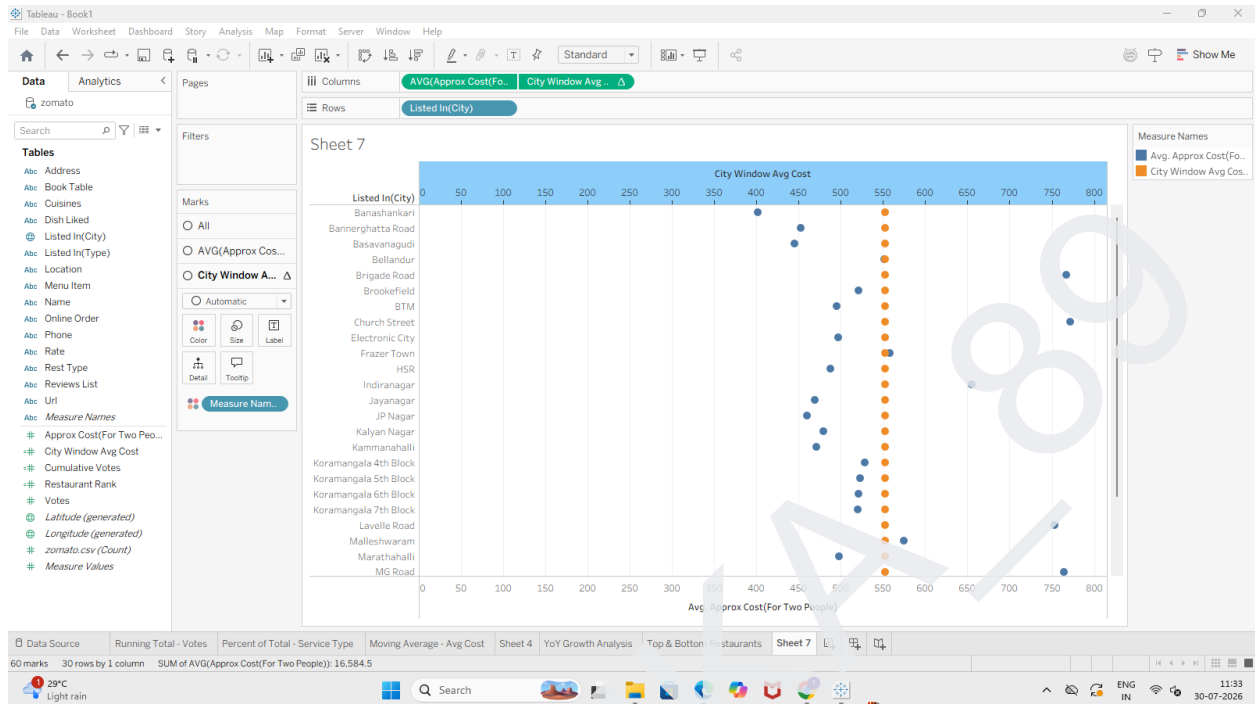
NAME : TANISHKA KUTWAL

ROLL NO:- T089

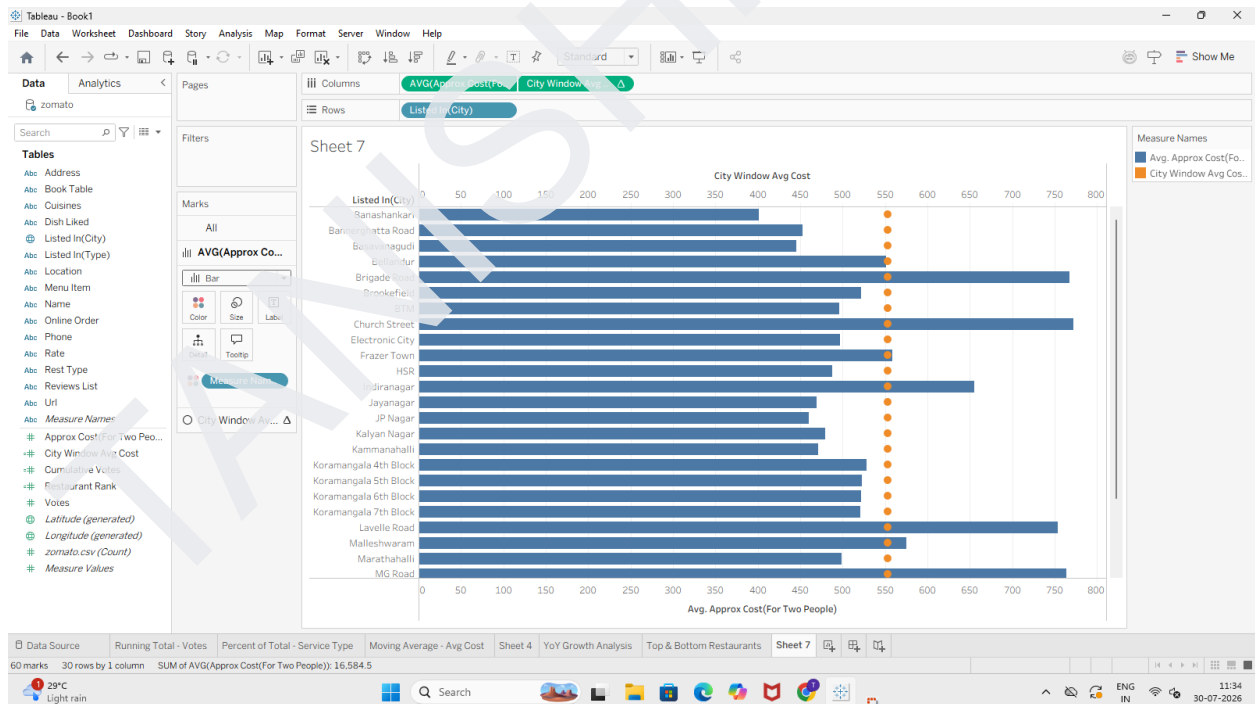
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SUB:-Data Visualization with Power BI and Tableau

PRACTICAL NO.:-5



Change the mark type for City Window Avg Cost to Line (or Gantt Bar).

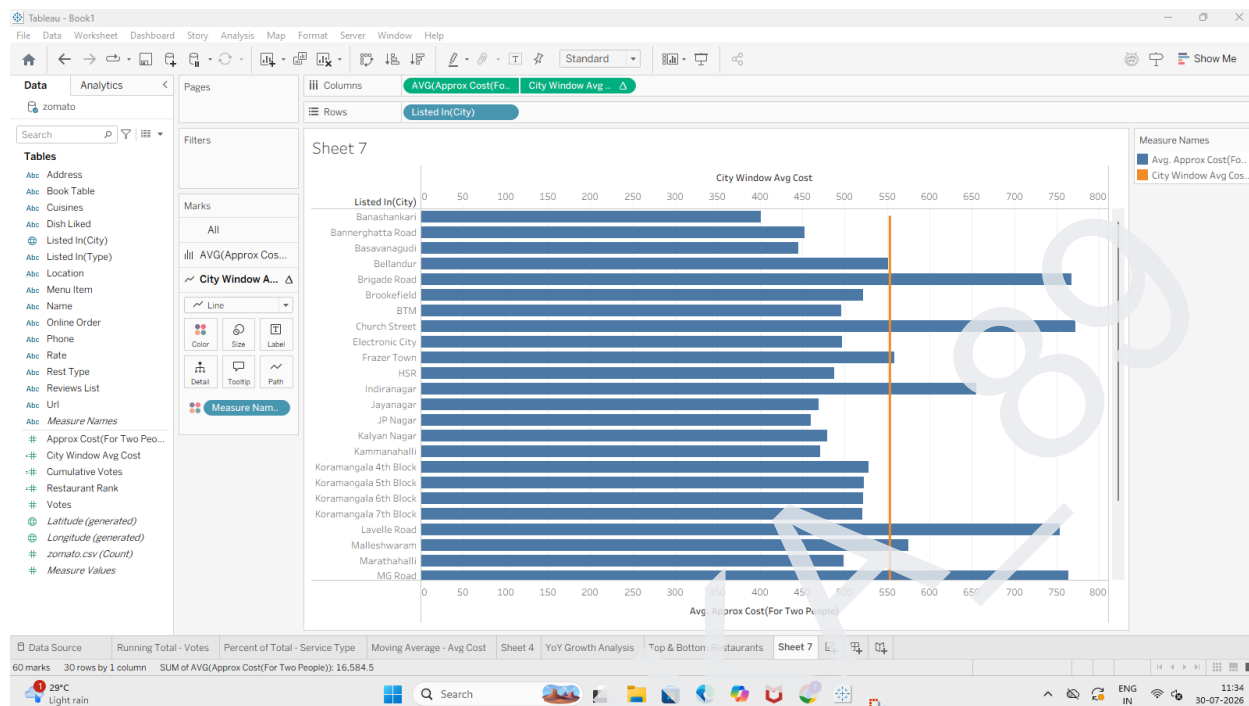


NAME : TANISHKA KUTWAL
ROLL NO:- T089

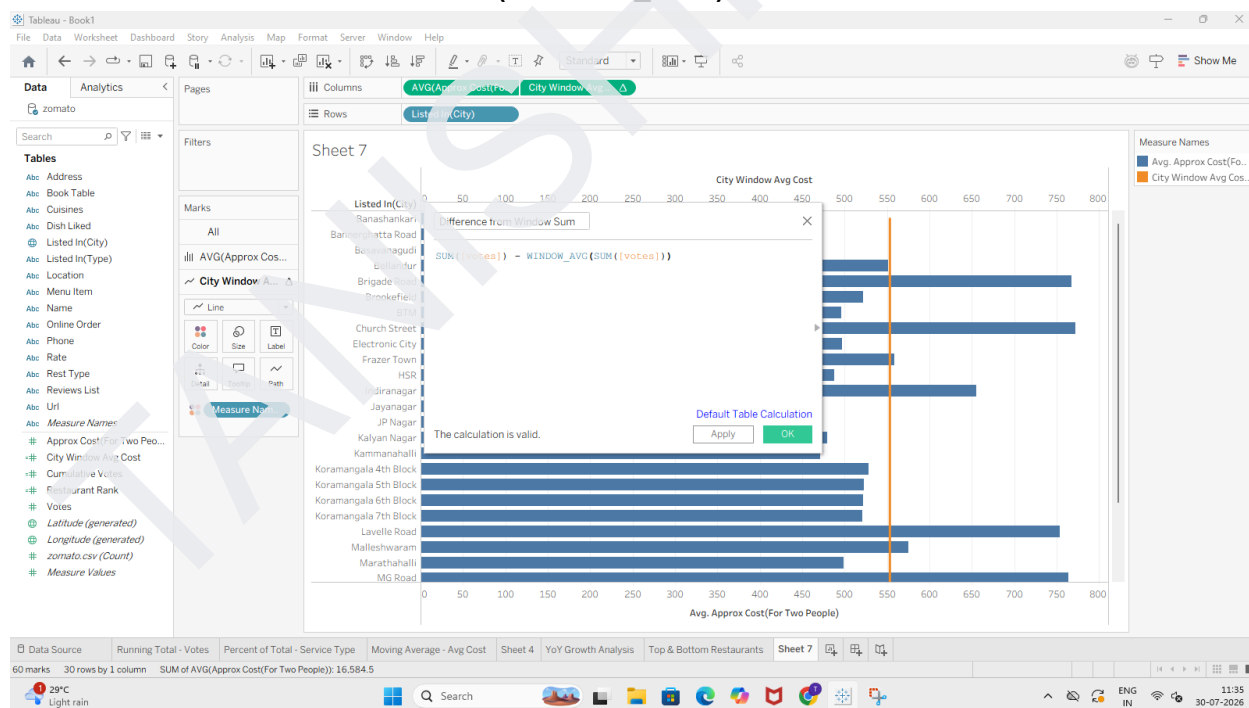
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PRACTICAL NO.:-5



Create a Window Difference Metric (WINDOW_SUM)



Open new Sheet

o Drag listed_in(city) to the Rows shelf and votes to the Columns shelf.

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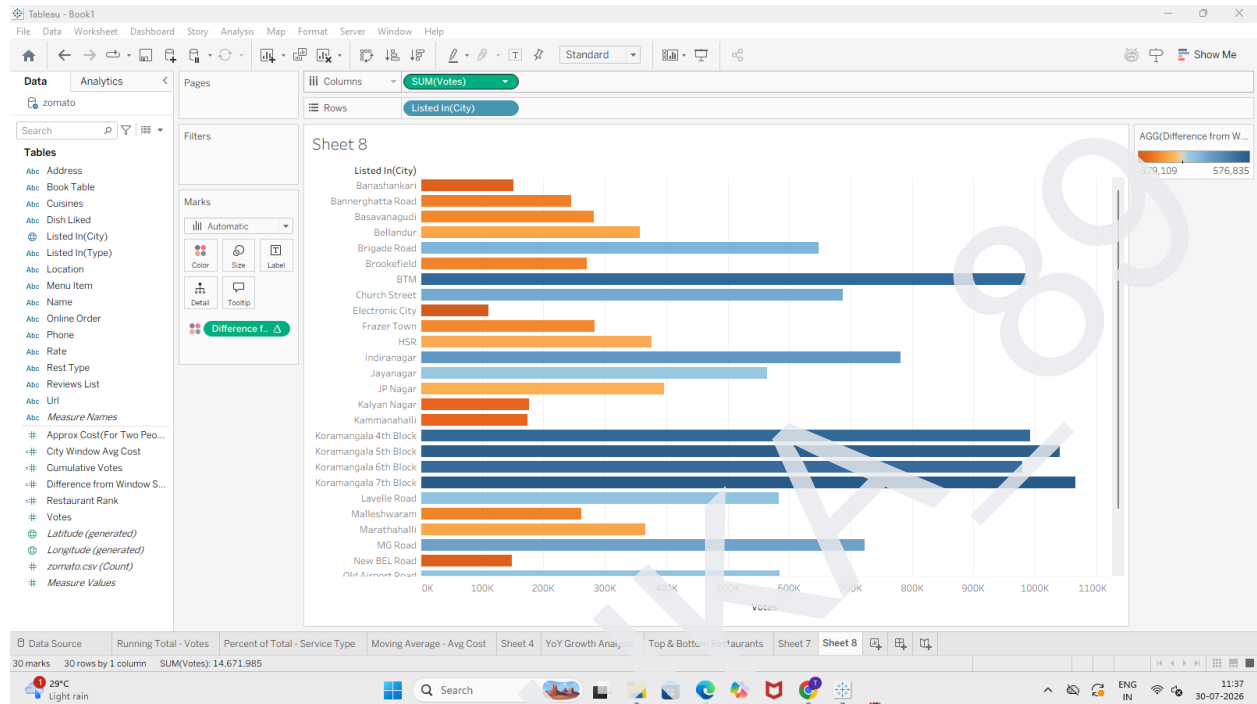
ROLL NO:- T089

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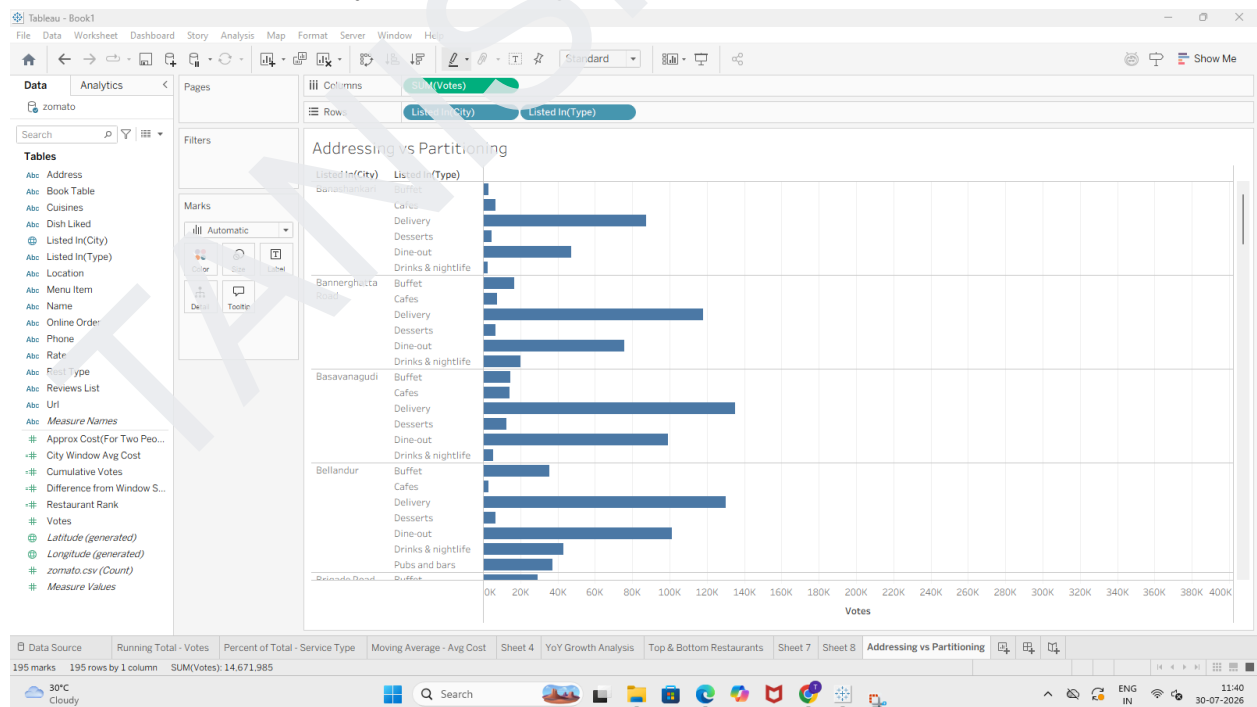
PRACTICAL NO.:-5

o Drag your new Difference from Window Sum calculated field onto the Color mark on the Marks card.



F: Customize addressing and partitioning

Build a Nested View for City and Service Type



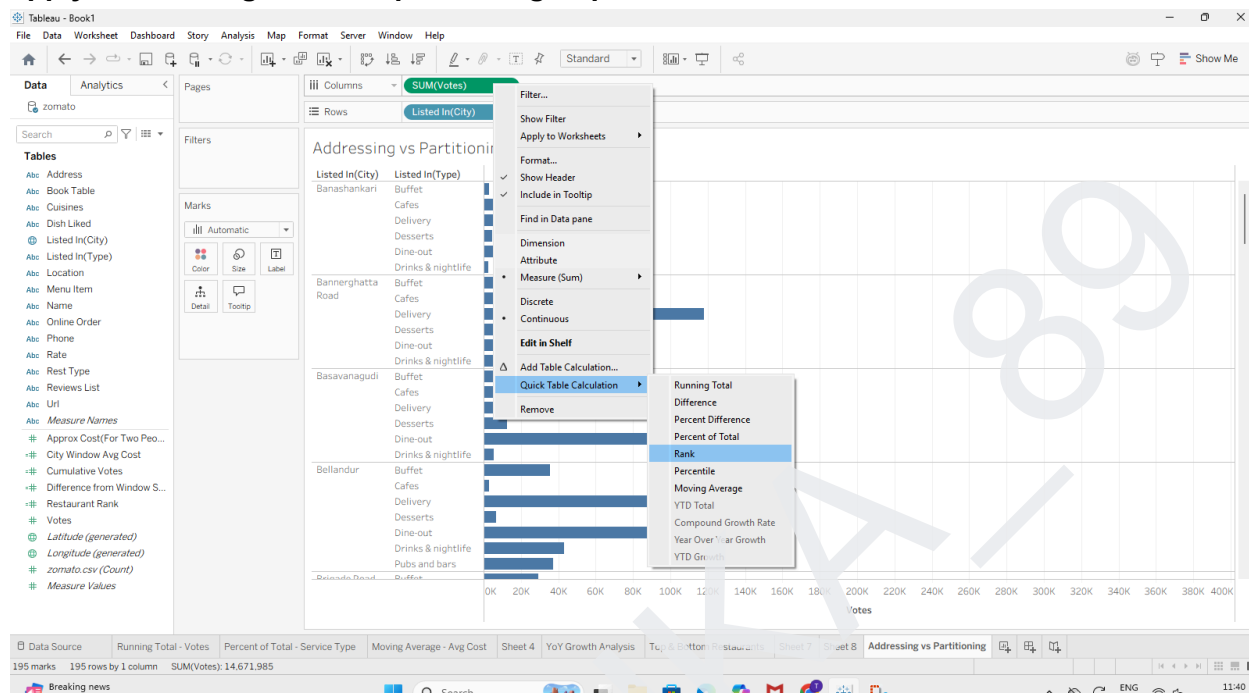
NAME : TANISHKA KUTWAL
ROLL NO:- T089

SHETH L.U.J. AND SIR M.V. COLLEGE

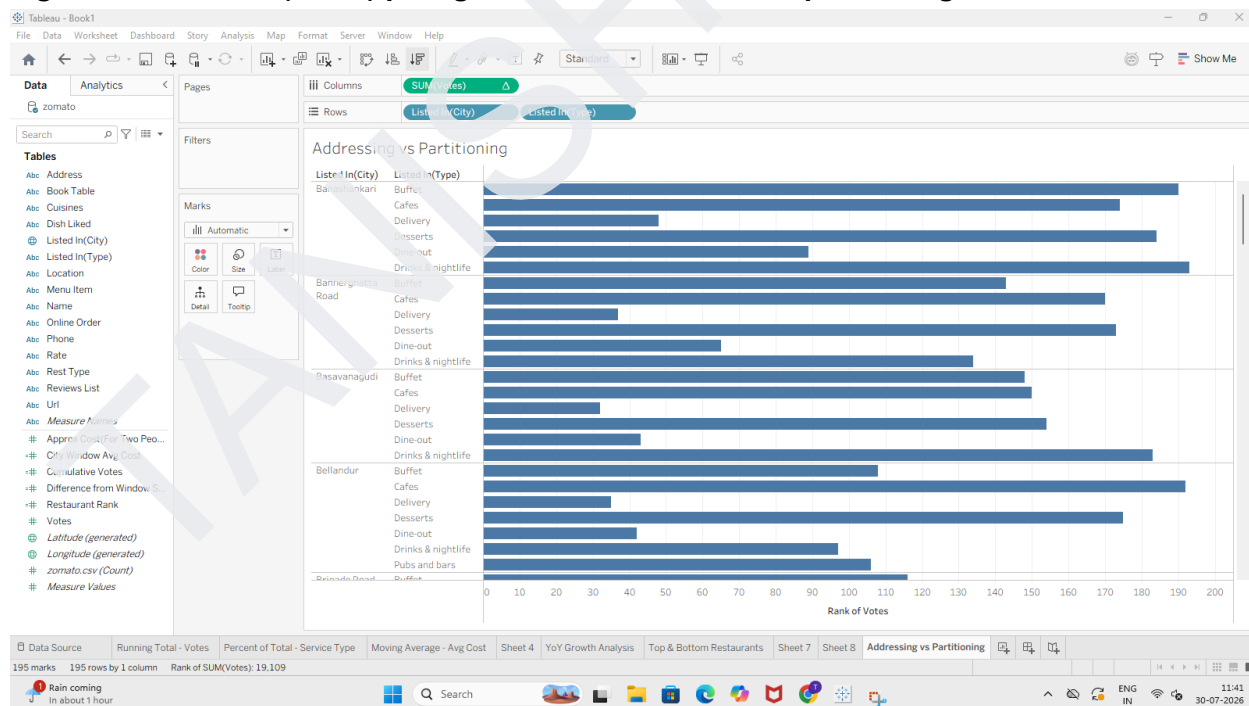
SUB:-Data Visualization with Power BI and Tableau

PRACTICAL NO.:-5

Apply Partitioning via "Compute Using" Options



Right-click the SUM(votes) pill again and hover over Compute Using:

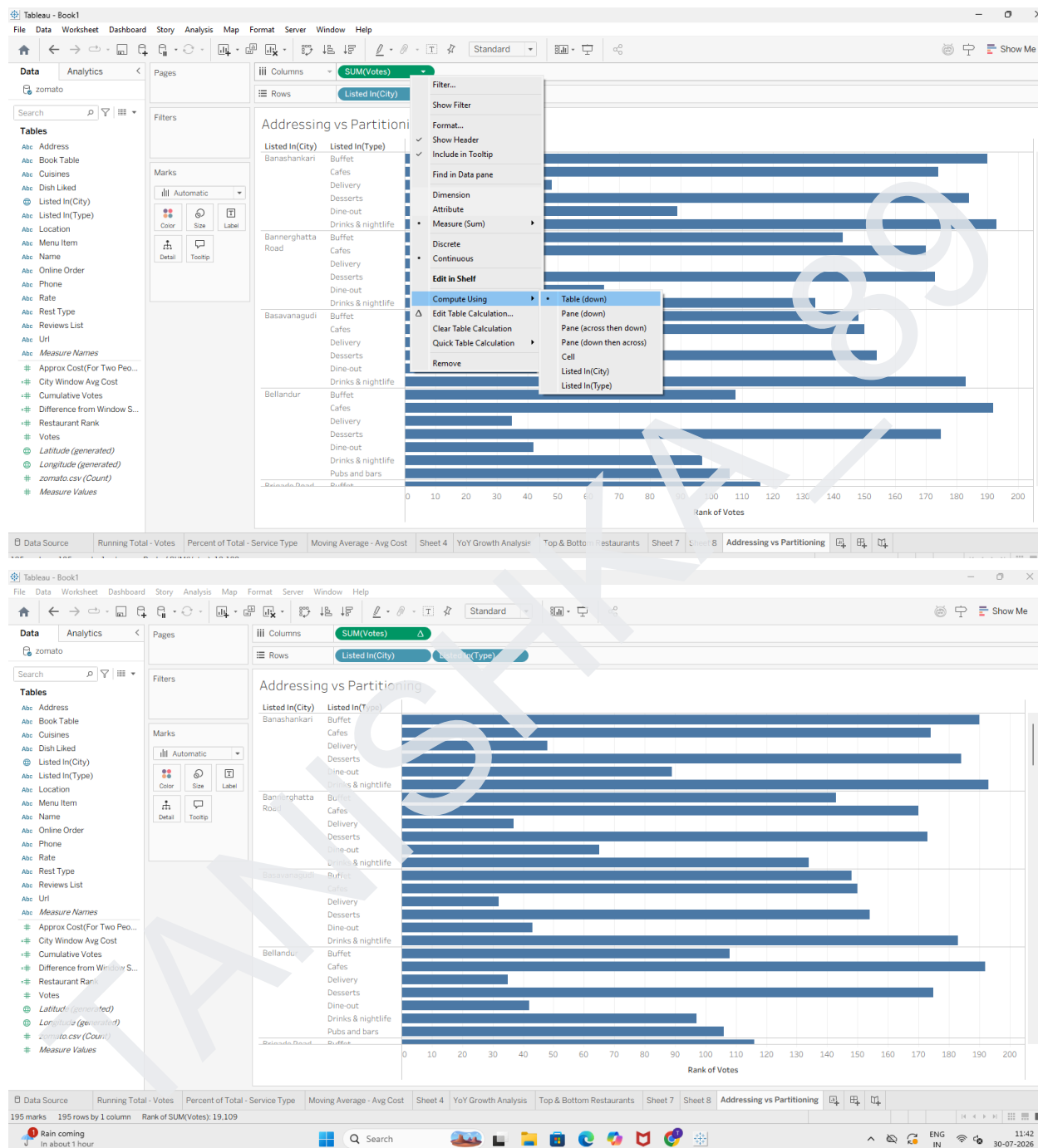


NAME : TANISHKA KUTWAL
ROLL NO:- T089

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SUB:-Data Visualization with Power BI and Tableau

PRACTICAL NO.:-5



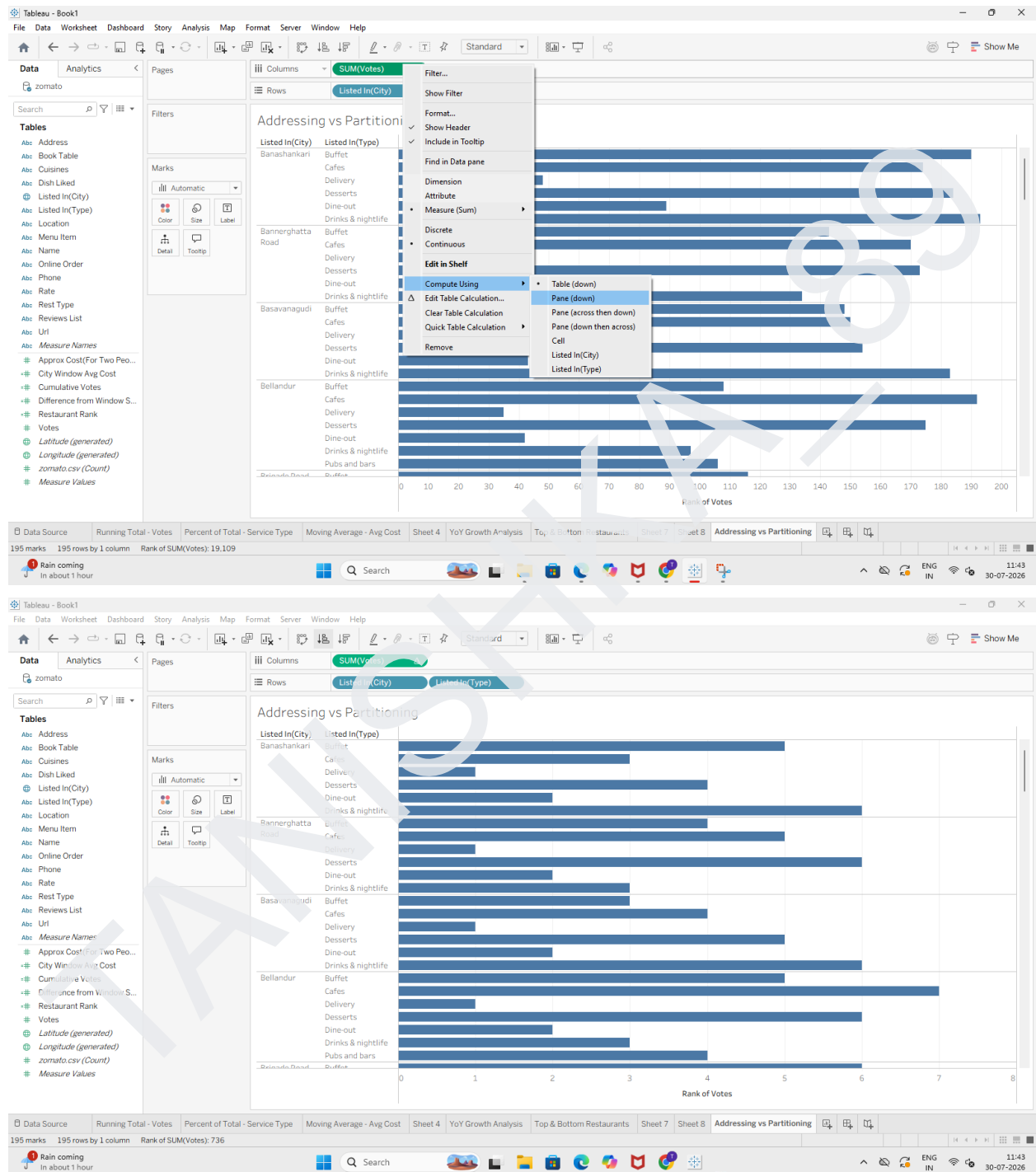
NAME : TANISHKA KUTWAL
ROLL NO:- T089

SHETH L.U.J. AND SIR M.V. COLLEGE

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PRACTICAL NO.:-5

Manually Define Addressing with Specific Dimensions

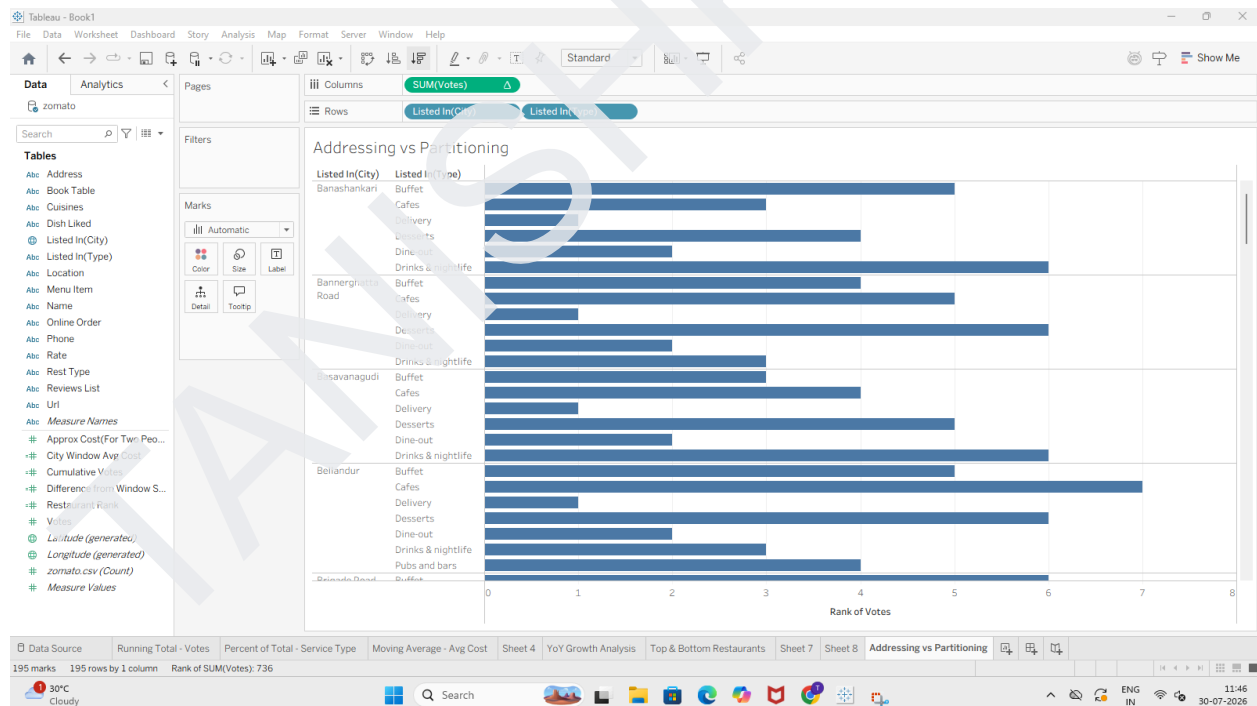
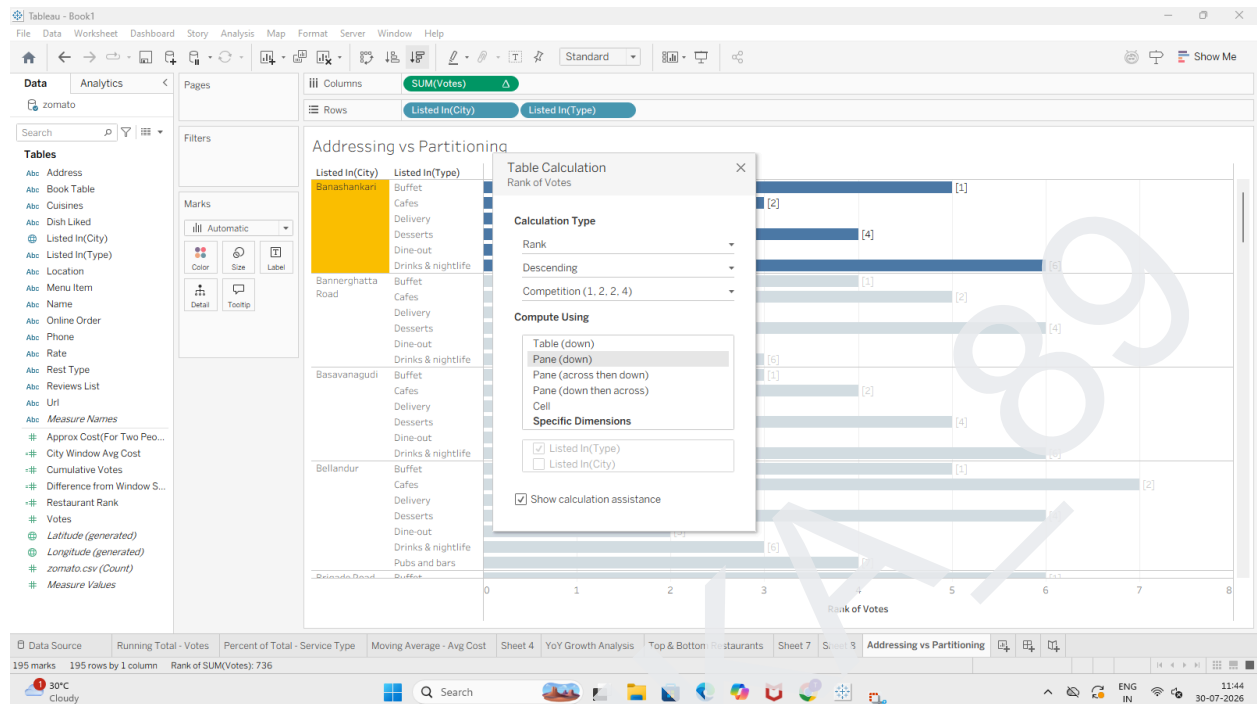


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ROLL NO:- T089

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G: Create dynamic titles and labels using table calculations

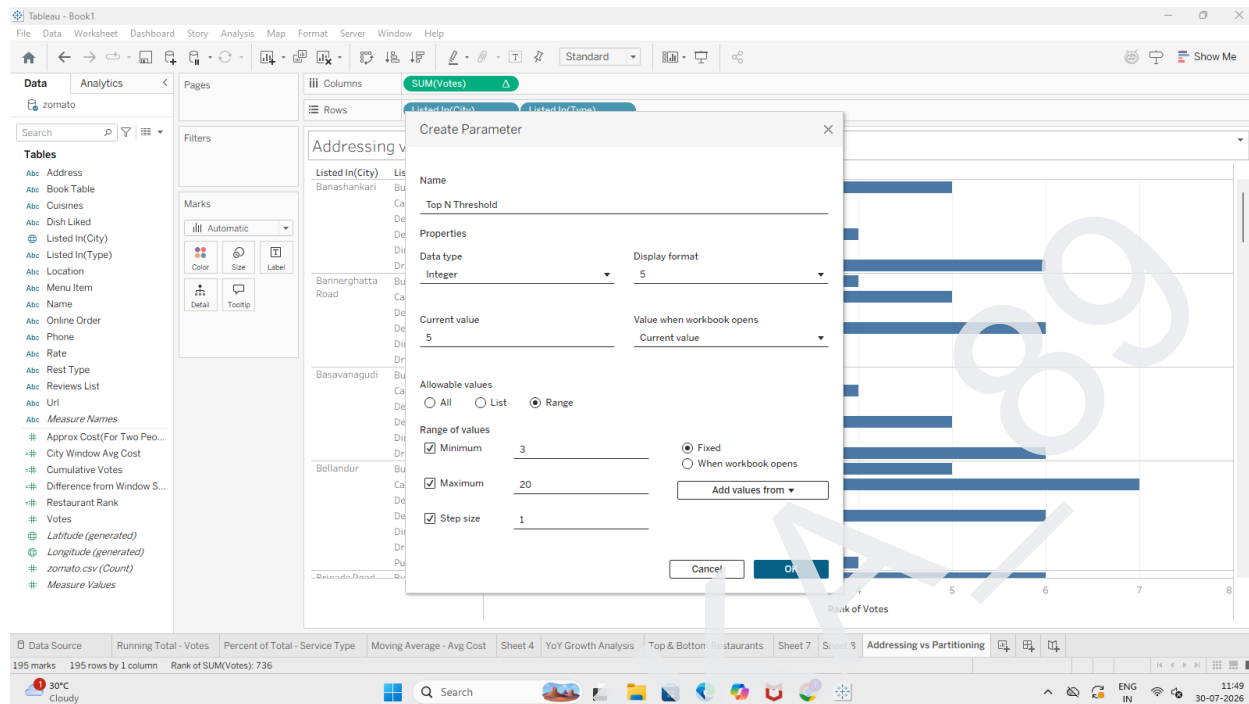
NAME : TANISHKA KUTWAL

ROLL NO:- T089

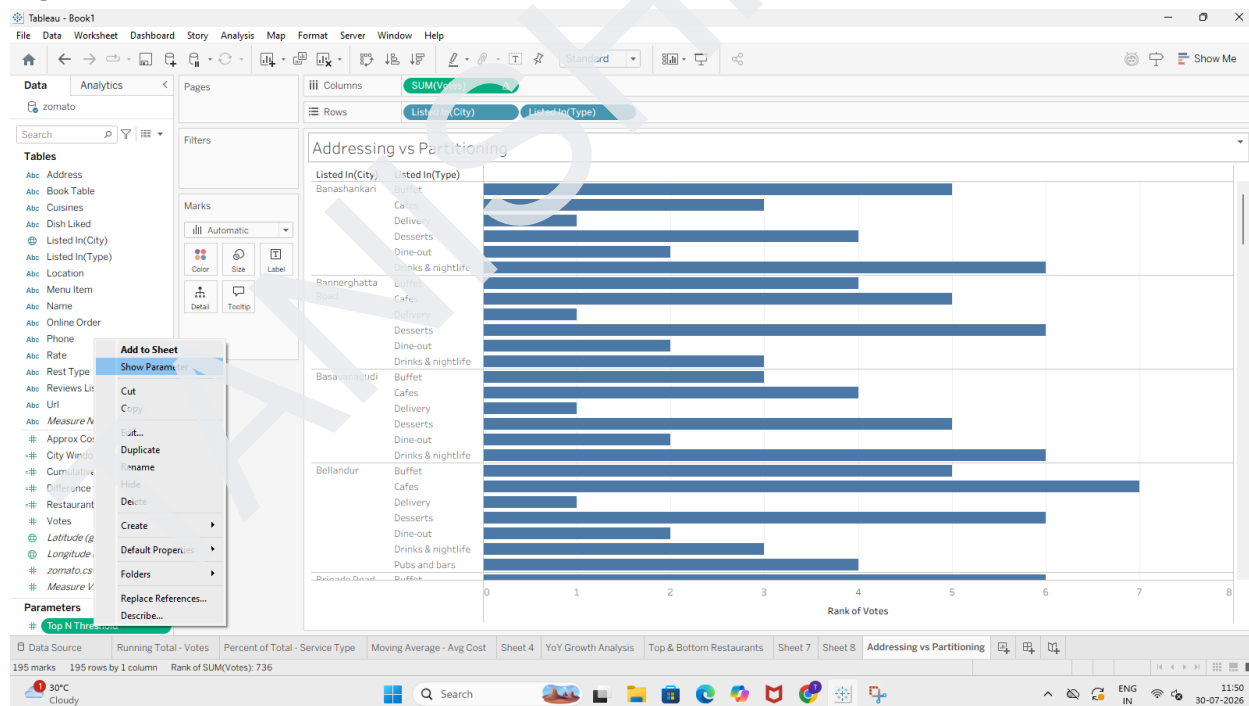
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Right-click Top N Threshold in the Data Pane and select Show Parameter Control.



Create a Dynamic Header Calculation

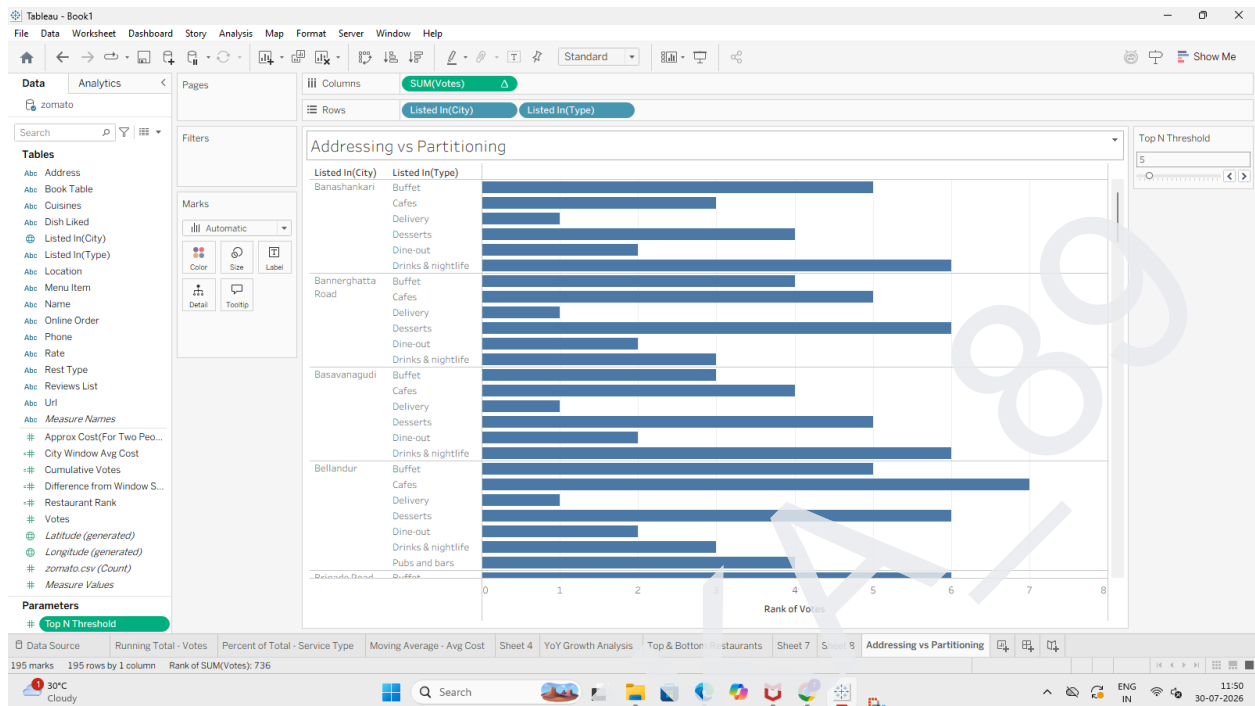
NAME : TANISHKA KUTWAL

ROLL NO.:- T089

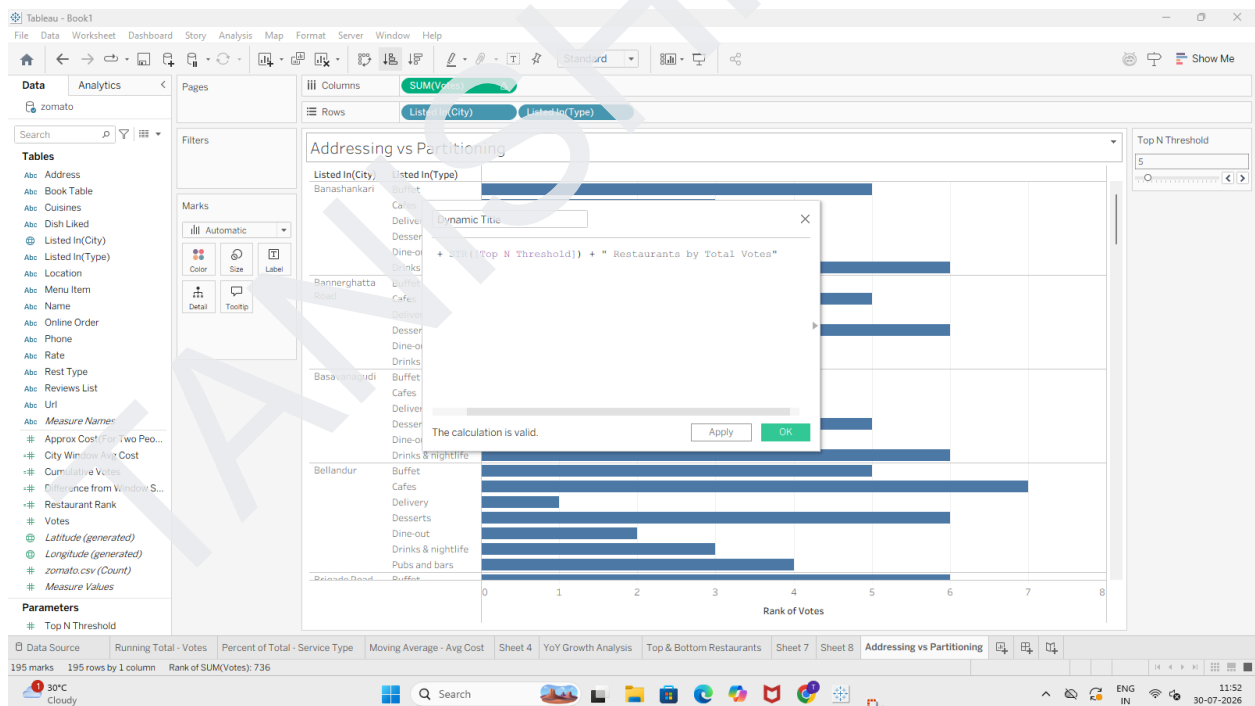
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Embed Calculated Fields and Parameters into the Sheet Title



In the Edit Title dialog box, click the Insert button in the top right corner.

NAME : TANISHKA KUTWAL
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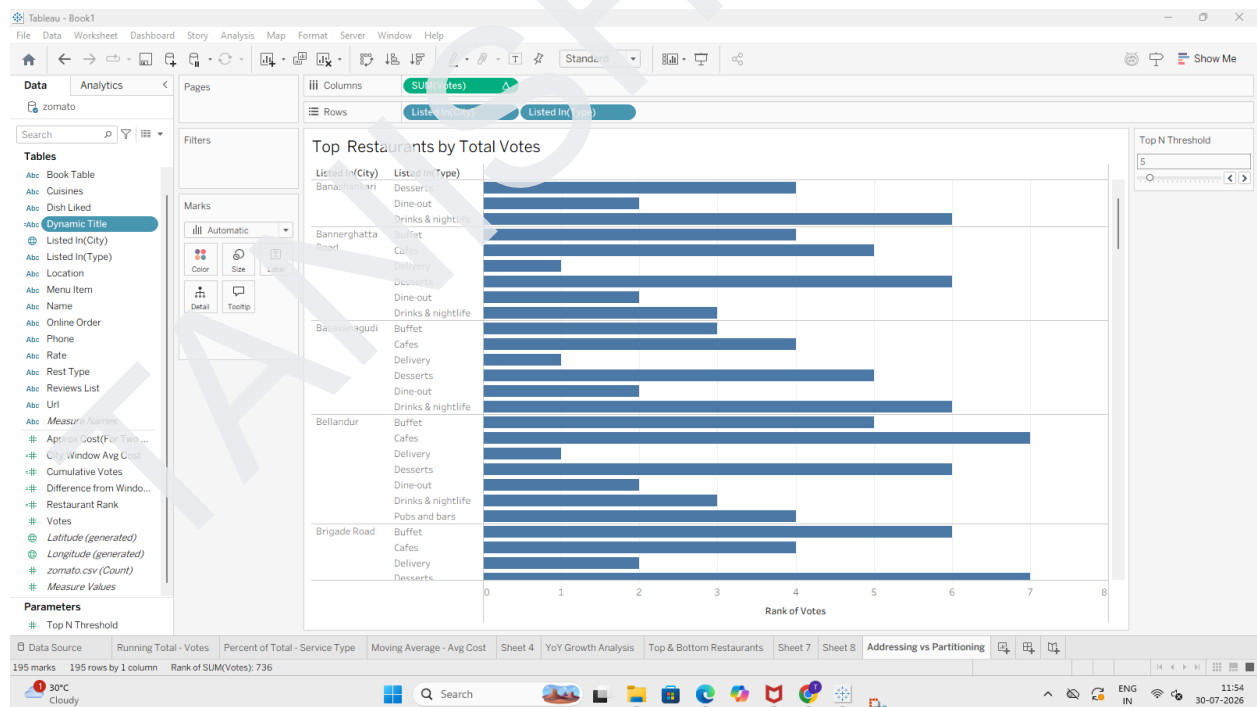
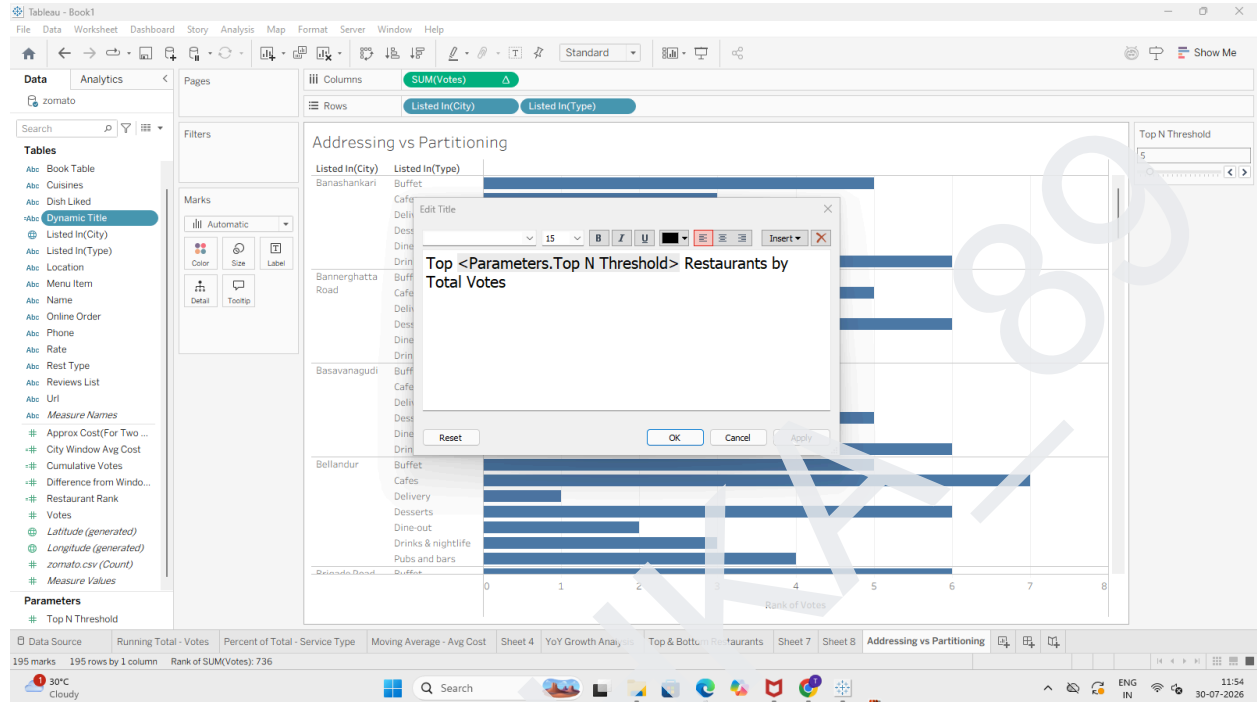
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Format the title text so it reads:

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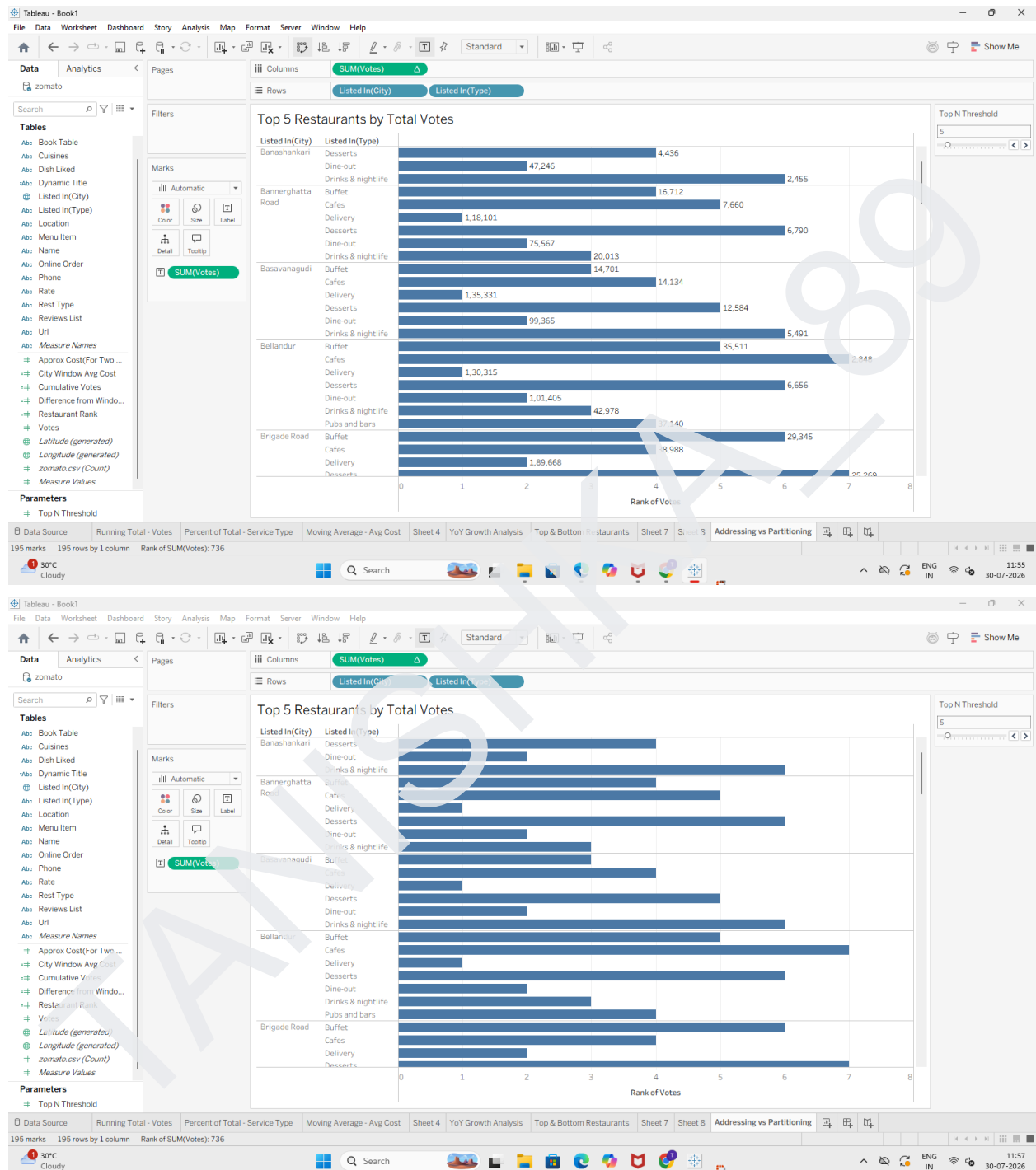


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